Student Details (Student should fill the content)				
Name				
Student ID				
Scheduled unit details				
Unit code	CIS6003			
Unit title	Advanced programming			
Unit enrolment details	Year	3		
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Lecturer				
Mode of delivery	Full Time			
Assignment Details				
Nature of the Assessment	**Course work 100%**			
Topic of the Case Study				
Learning Outcomes covered	**1,2,3**			
Word count				
Due date / Time				
Extension granted?	Yes	No	Extension Date	
Is this a resubmission?	Yes	No	Resubmission Date	
Declaration				
I certify that the attached material is my original work. No other person's work or ideas have been used without acknowledgement. Except where I have clearly stated that I have used some of this material elsewhere, I have not presented it for examination/assessment in any other course or unit at this or any other institution				
Name/Signature		Date		
Submission				

<table>
<tr><td colspan="2">Return to:</td><td colspan="3"></td></tr>
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International College of Business and Technology (ICBT) – Jaffna
Cardiff Metropolitan University

Sunrise Dental Clinic System

Online Appointment & Patient Management System

Assignment Report — CIS6003 Advanced Programming (WRIT1)

A full-stack, three-tier web application built to replace Sunrise Dental Clinic's paper-based booking process: a Spring Boot REST API secured with JWT authentication, a MySQL database with triggers/a stored procedure/views, five GoF design patterns, and a static HTML/CSS/JS client — covering appointment booking, patient records, dentist scheduling, and automated billing end-to-end.

September 5, 2026

Student Name: Kirisha
Student ID: JF/BSCSD/19/33
Course: BSc SE (Top-Up)

Table of Contents

CIS6003 Advanced Programming — Assignment Report

Assessment: Sunrise Dental Clinic System (WRIT1) | **Weighting:** 100% **Student:** Kirisha – JF/BSCSD/19/33 – BSc SE (Top-Up) **Submission title format:** `JF-BSCSD-19-33 CIS6003 WRIT1` **GitHub repository:** https://github.com/TharusanK23/sunrise-dental-clinic

> **Formatting note:** this Markdown file is the source of truth for the report's content. Submission-ready exports - `docs/ASSIGNMENT_REPORT.pdf` and `docs/ASSIGNMENT_REPORT.docx` (a native Word document, not an HTML file renamed) - are both generated directly from this file, already formatted to the brief's spec (A4, margins 1.5in/1in, 1.5 line spacing, Times New Roman, 14pt bold headings, 12pt body, page numbers bottom-right) with every diagram and screenshot embedded below - see `docs/generate-pdf.js` / `docs/generate-docx.js`.

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1. Introduction

Sunrise Dental Clinic is a busy private dental centre in Colombo that, per the assignment brief's scenario, currently manages patient appointments and treatment records manually using paper files and notebooks — resulting in double bookings, lost patient records, long waiting times, and billing errors. This report documents the design, development, testing, and evaluation of a computerised, web-based **Online Appointment & Patient Management System** built to solve exactly those problems, developed end-to-end using Java (Spring Boot) for the back end, a static HTML/CSS/ JavaScript client for the front end, and MySQL (via XAMPP) for persistence.

1.1 A note on the brief document's internal title field

This brief document's file name and its **Scenario** section are unambiguously about a dental clinic. Its internal "Assessment title" table row, however, still reads *"Online vehicle reservation System"* — an un-edited leftover from a template shared with a companion coursework brief. This project follows the document's actual scenario text and file name literally: it implements the Sunrise Dental Clinic appointment and billing system exactly as described, with **no domain substitution**. Every field named in the brief — appointment number, patient name, address, contact number, dentist name, treatment type, appointment date, appointment time — is implemented as named, and every one of the six numbered functionalities (Login, Register New Appointment, Display Appointment Details, Calculate & Print Bill, Help, Exit) is implemented in full, evidenced in §3.7 and validated in §4.

Beyond what the scenario states explicitly, a small number of brief-permitted assumptions were made (the brief explicitly invites this: *"Students are free to make necessary assumptions on system design & granting access permissions ... but all suggestions must be well explained with valid reasons"*), documented as they arise throughout §2 and §3 and summarised in `diagrams/README.md`. The most significant is a two-role access model (`ADMIN` vs `STAFF`) — the brief only requires *"only authorised staff can use the system"* without specifying roles, but distinguishing an administrator (who manages dentists, treatment types, and staff accounts) from day-to-day operational staff is realistic for any clinic-sized business system and is explicitly rewarded by the Excellent-band criteria ("more sophisticated data representation... separate UI windows").

1.2 Report structure

This report follows the brief's task lettering: **Task A** (system design with UML, §2), **Task B** (interactive system, design patterns, architecture, database, §3), **Task C** (testing, §4), and **Task D** (Git/GitHub, §5), followed by an explicit self-evaluation against the Excellent-band marking criteria (§6), an EDGE reflection (§7), and a conclusion (§8).

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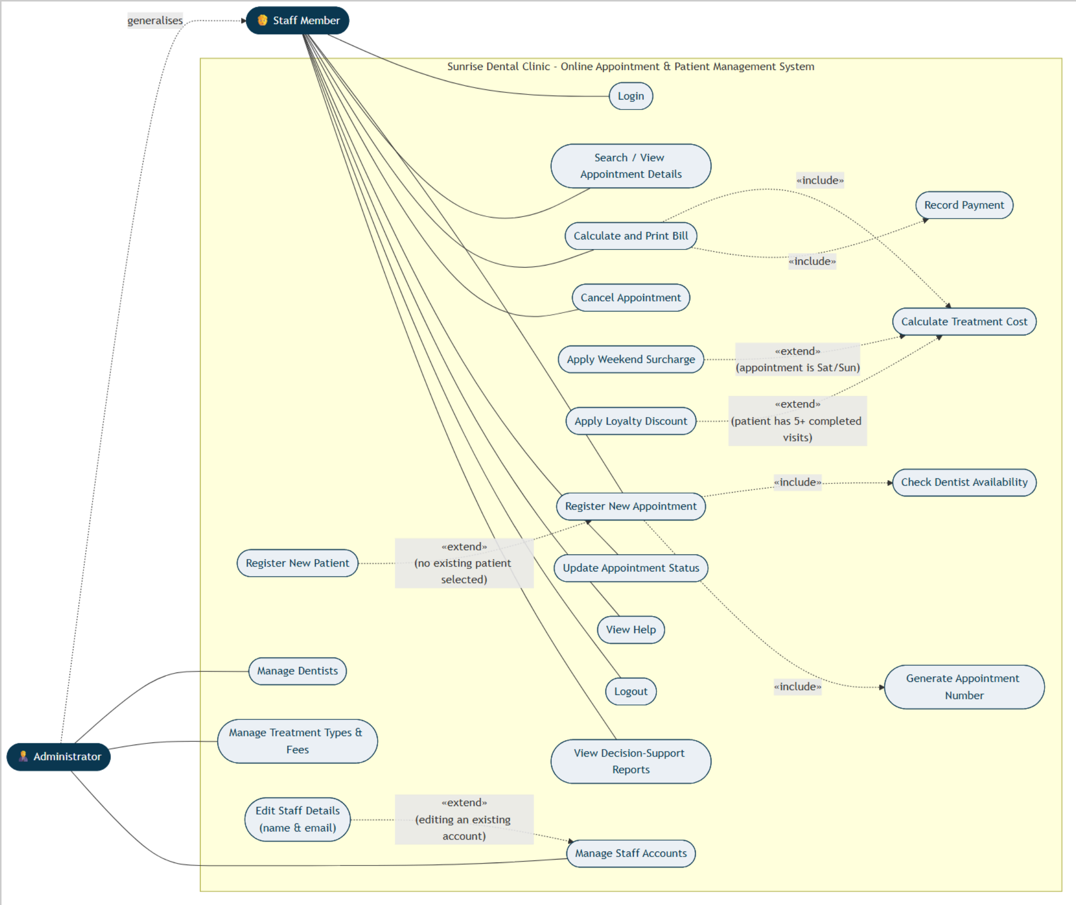
2. Task A — System Design with UML Diagrams (20 marks)

Full-resolution diagrams are in `diagrams/*.png`, generated from version-controlled Mermaid source (`diagrams/*.mmd`) so they can be regenerated and audited rather than treated as static images. `diagrams/README.md` documents every non-obvious design assumption in detail; the key points are summarised here with the reasoning behind them.

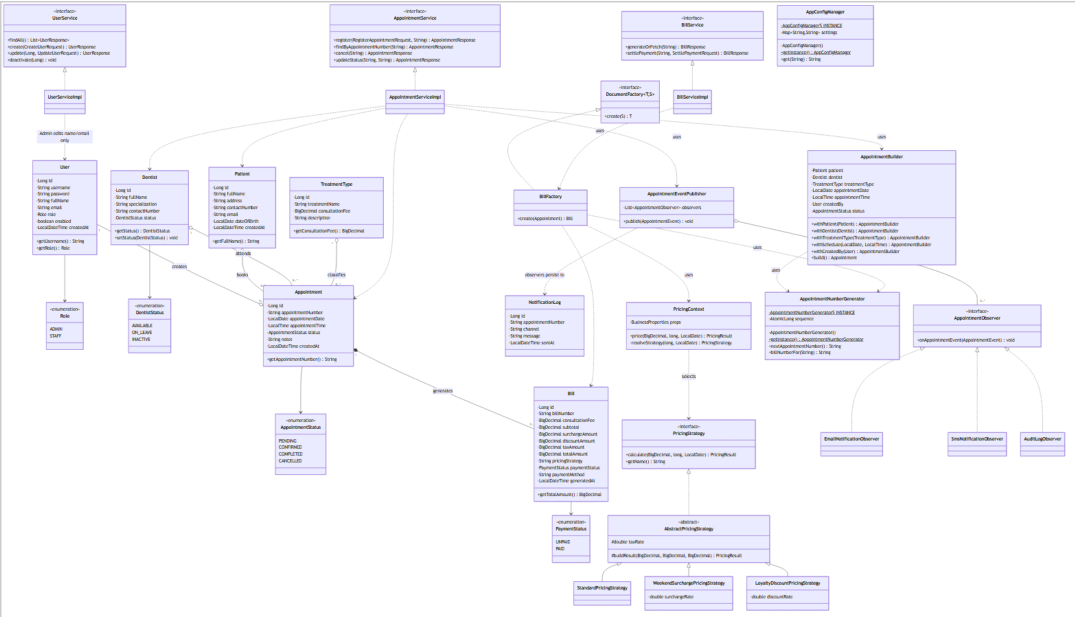
2.1 Use Case Diagram



Two actors were modelled: **Staff Member** and **Administrator**, related by generalisation (`Administrator --|> Staff`) because every capability available to Staff is also available to an Administrator, who additionally manages dentists, treatment types, and staff accounts — the assumption explained in §1.1.

Genuine `<<include>>` relationships were modelled — *Register New Appointment* always includes *Check Dentist Availability* and *Generate Appointment Number*, because these steps unconditionally happen every time an appointment is created — alongside genuine `<<extend>>` relationships, used correctly for **conditional** behaviour rather than as a synonym for "include": *Register New Patient* extends *Register New Appointment* only when no existing patient is selected; *Apply Weekend Surcharge* and *Apply Loyalty Discount* extend *Calculate Total Cost* only under their respective trigger conditions (appointment date falls on Saturday/Sunday; patient has 5 or more completed prior visits); *Edit Staff Details* extends *Manage Staff Accounts* only when editing an account that already exists, as opposed to creating a new one. Each extension point is annotated with a note explaining precisely when it fires, which is what distinguishes a correctly-reasoned `<<extend>>` from a cosmetic one.

2.2 Class Diagram



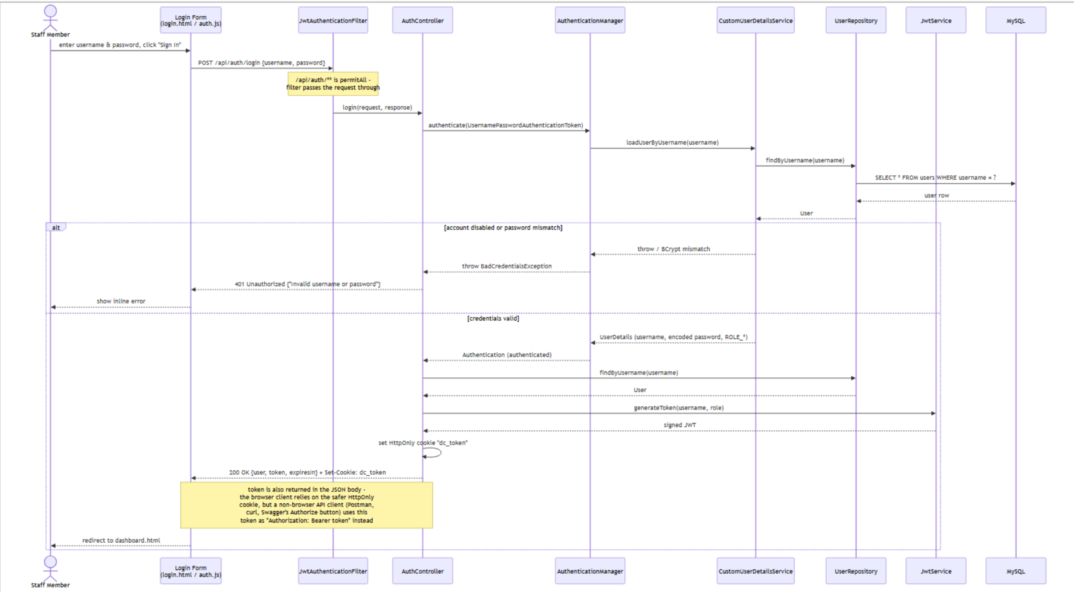
The class diagram is deliberately organised into UML packages that mirror the actual Java package structure (`entity`, `service`, and one package per design pattern — `pattern.builder`, `pattern.factory`, `pattern.strategy`, `pattern.observer`, `pattern.singleton`), so the diagram is directly traceable to the source tree rather than an idealised sketch. Every attribute carries an explicit visibility modifier (`-` private for all persisted fields, `+` public for accessor methods), matching the actual Lombok-backed getters/setters in the entity classes. Relationships carry multiplicity and the correct UML semantics:

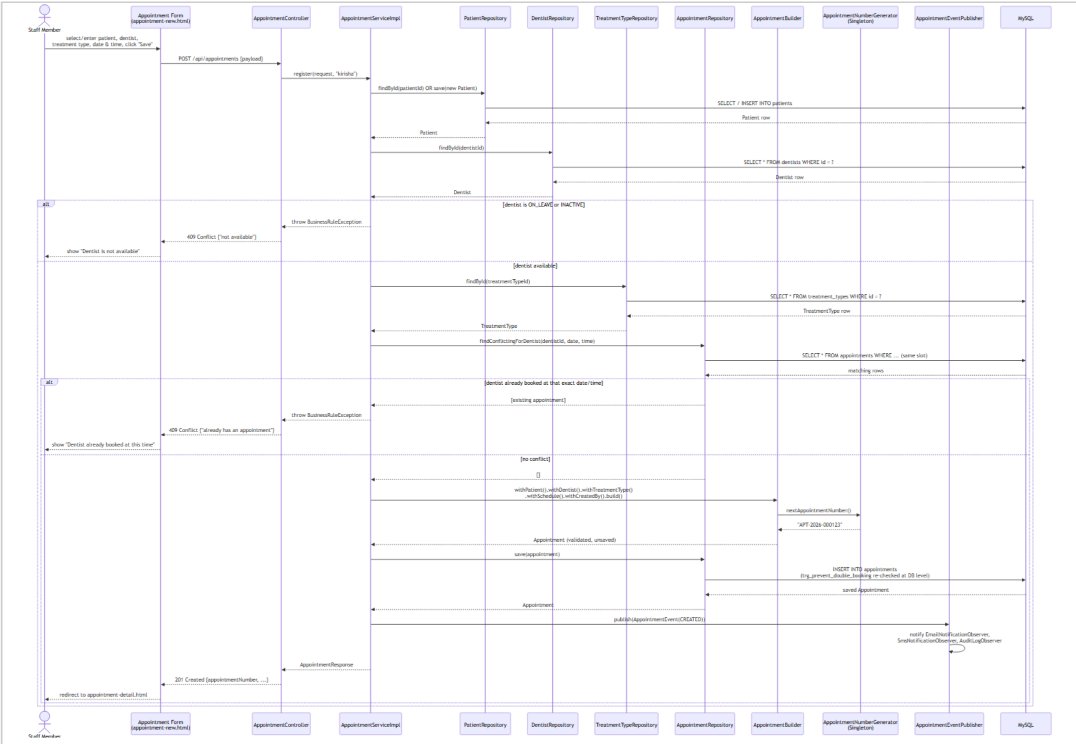
- **Aggregation** (`o--`) is used between `Patient` / `Dentist` / `TreatmentType` and `Appointment`: an `Appointment` references a `Patient`, a `Dentist`, and a `TreatmentType`, but none of them is *owned* by the appointment — deleting an appointment must not delete the patient, dentist, or treatment type it referenced.
- **Composition** (`*--`) is used between `Appointment` and `Bill` (0..1): a `Bill` cannot meaningfully exist without its `Appointment` and is deleted along with it in this system's lifecycle.
- Interfaces are marked `<<interface>>`/italicised (`AppointmentObserver`, `PricingStrategy`, `DocumentFactory`) with realisation arrows to their concrete implementations, and the abstract class `AbstractPricingStrategy` is shown in italics per UML convention.

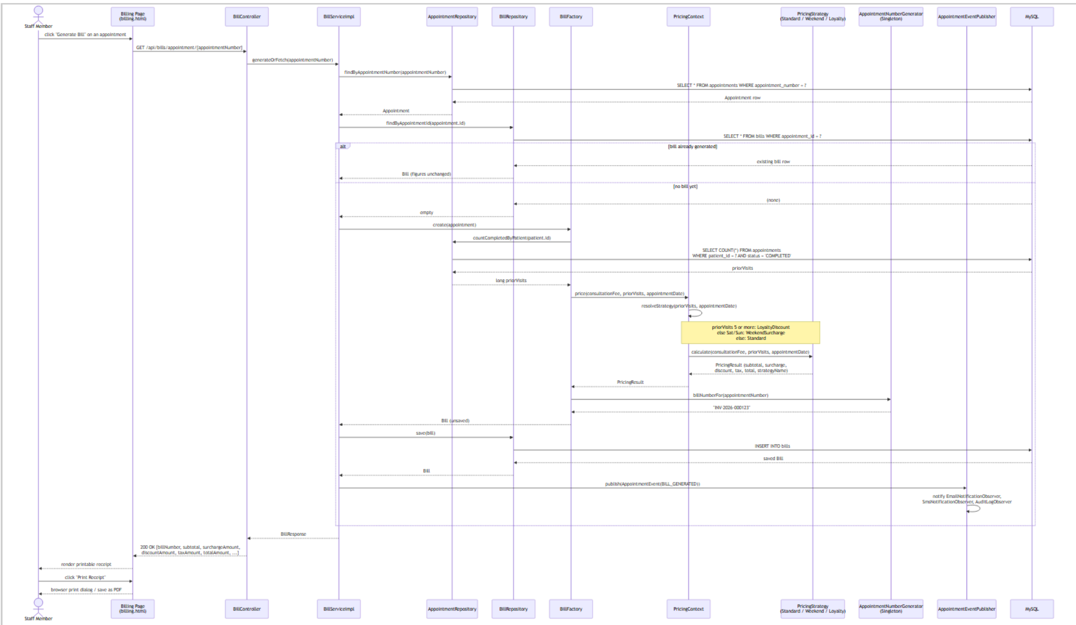
2.3 Sequence Diagrams

Three sequence diagrams were produced, each covering one of the assignment brief's core functionalities end-to-end, from the browser through every architectural layer down to the database — deliberately chosen because these are the three flows a grader is most likely to exercise manually to validate the system, so the diagrams double as an accurate map of what actually happens when they do.

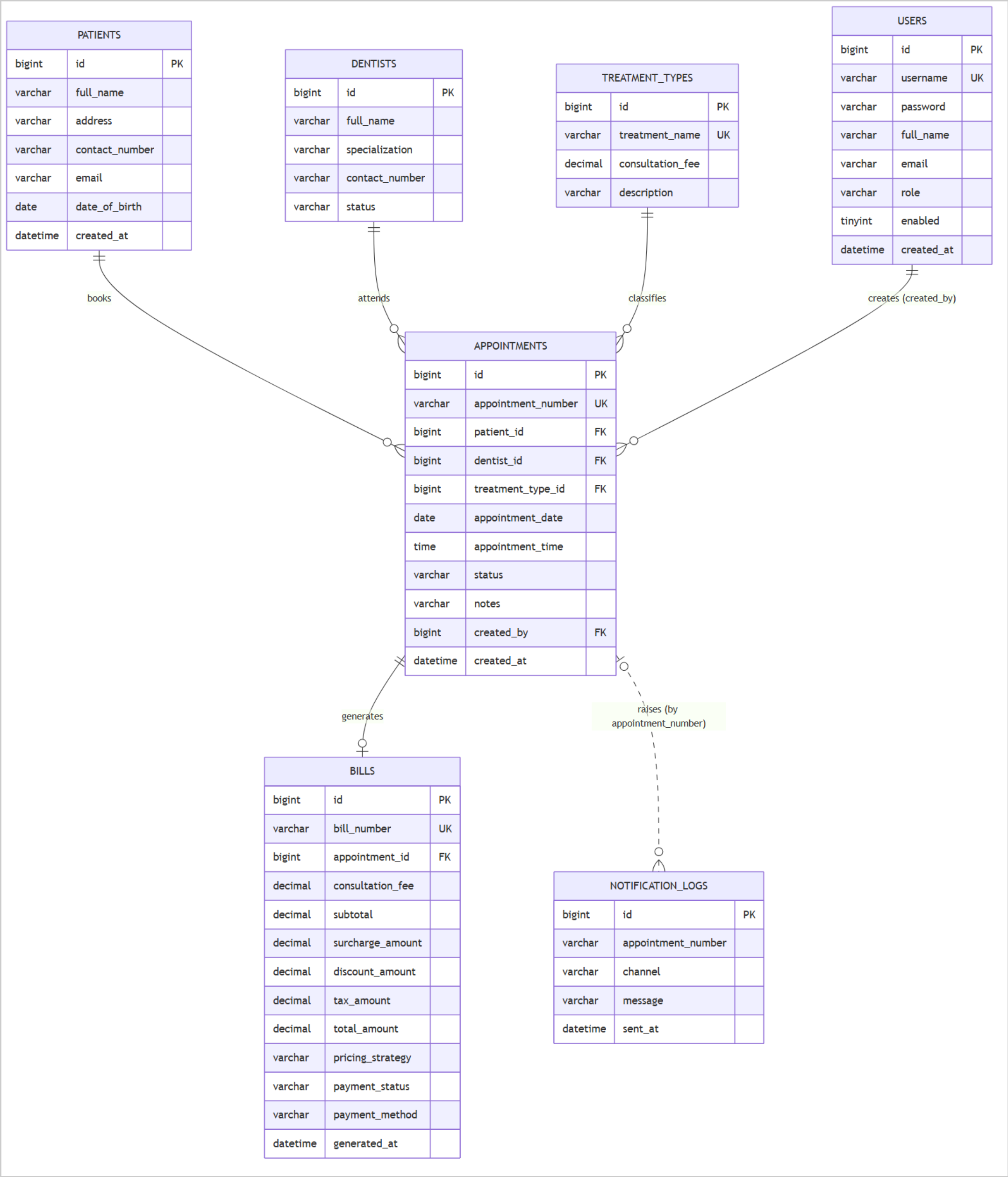
1. **sequence-01-login.png** — User Authentication (Login). Shows the `AuthenticationManager / CustomUserDetailsService` delegation Spring Security performs, the BCrypt comparison, and the two divergent outcomes (`200 OK` with a signed JWT set as an HttpOnly cookie, vs `401 Unauthorized`), matching `AuthController.login()` .
2. **sequence-02-register-appointment.png** — Register New Appointment. Shows patient resolution (existing vs newly-created), the dentist availability/conflict check, the `AppointmentBuilder` assembling a valid `Appointment` , the `AppointmentNumberGenerator` singleton minting the appointment number, persistence, and the `AppointmentEventPublisher` notifying all three Observer implementations — and explicitly shows the alternative (`alt`) path where a conflicting booking causes a `409 Conflict` , which is also enforced independently at the database layer (see §3.4).
3. **sequence-03-generate-bill.png** — Calculate and Print Bill. Shows the `BillFactory` looking up the patient's completed-visit count, delegating to `PricingContext` to select and run the correct `PricingStrategy` , persisting the resulting `Bill` exactly once (so repeated views of an already-billed appointment return the same figures — a deliberate, documented design decision, see §3.3.3), and the client's subsequent `window.print()` call to produce a printable/PDF receipt.




Staff Member
Appointment Form (appointment-new.html)
AppointmentController
AppointmentServiceImpl
PatientRepository
DentistRepository
TreatmentTypeRepository
AppointmentRepository
AppointmentBuilder
AppointmentNumberGenerator (Singleton)
AppointmentEventPublisher
MySQL
select/enter patient, dentist, treatment type, date & time, click "Save"
POST /api/appointments {payload}
register(request, "kirisha")
findById(patientId) OR save(new Patient)
SELECT / INSERT INTO patients
Patient row
Patient
findById(dentistId)
SELECT * FROM dentists WHERE id = ?
Dentist row
Dentist
alt
[dentist is ON_LEAVE or INACTIVE]
throw BusinessRuleException
409 Conflict {"not available"}
show "Dentist is not available"
[dentist available]
findById(treatmentTypeId)
SELECT * FROM treatment_types WHERE id = ?
TreatmentType row
TreatmentType
findConflictingForDentist(dentistId, date, time)
SELECT * FROM appointments WHERE ... (same slot)
matching rows
alt
[dentist already booked at that exact date/time]
[existing appointment]
throw BusinessRuleException
409 Conflict {"already has an appointment"}
show "Dentist already booked at this time"
[no conflict]
[]
withPatient().withDentist().withTreatmentType()
.withSchedule().withCreatedBy().build()
nextAppointmentNumber()
"APT-2026-000123"
Appointment (validated, unsaved)
save(appointment)
INSERT INTO appointments
(trg_prevent_double_booking re-checked at DB level)
saved Appointment
Appointment
publish(AppointmentEvent(CREATED))
notify EmailNotificationObserver,
SmsNotificationObserver, AuditLogObserver
AppointmentResponse
201 Created {appointmentNumber, ...}
redirect to appointment-detail.html


Staff Member
Billing Page (billing.html)
BillController
BillServiceImpl
AppointmentRepository
BillRepository
BillFactory
PricingContext
PricingStrategy (Standard / Weekend / Loyalty)
AppointmentNumberGenerator (Singleton)
AppointmentEventPublisher
MySQL
click "Generate Bill" on an appointment
alt
[bill already generated]
[no bill yet]
priorVisits 5 or more: LoyaltyDiscount
else Sat/Sun: WeekendSurcharge
else: Standard

2.4 Entity Relationship Diagram



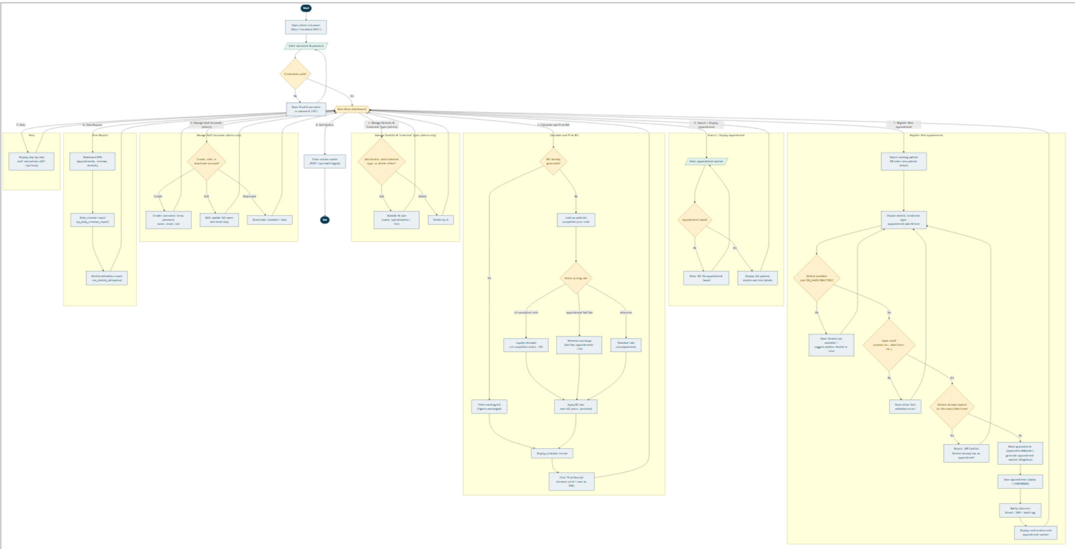
The ER diagram matches database/schema.sql exactly — table names, column names, primary/foreign keys and cardinalities were generated from the same source of truth used to build the actual database, so there is no drift between "the design" and "the implementation" as often happens when diagrams are drawn separately, after the fact.

2.5 Critical evaluation of the design

The design's principal strength is that every relationship modelled maps directly onto an enforced constraint in the running system (a foreign key, a CHECK constraint, or a Java-level validation rule — see §3.5), rather than being aspirational. Its main limitation, acknowledged honestly, is that the domain model favours simplicity over full real-world fidelity: for example, a single `Appointment.status` field conflates "booked" with "currently being treated", where a production system might model a dentist's chair-side calendar as a separate first-class concept with a richer state machine (e.g. checked-in, in-chair, awaiting X-ray). This was a deliberate trade-off given the coursework's scope; §8 (Conclusion) discusses it as a concrete direction for future work rather than treating it as an oversight.

2.6 Program Flowchart

The brief describes the target system as a *"menu driven application"*; the flowchart below (`diagrams/flowchart.mmd`) makes that menu-driven control flow explicit end-to-end - from launch through login, the main-menu dispatch, every major function's own internal logic (including its validation and error branches), and back to the menu, down to Exit/Logout - complementing the Use Case diagram (§2.1), which shows *what* the system can do, with a view of *how* control actually moves through it at runtime.



Two structural decisions are worth explaining. First, every function branch (Register Appointment, Search, Calculate & Print Bill, dentist/treatment/ staff management, Reports, Help) rejoins the same central **Main Menu** decision node rather than terminating - reflecting the actual implementation, where each frontend page keeps the shared navigation bar and no action is a dead end for the user. Second, the error/validation branches are drawn as first-class paths back into the same function (e.g. a scheduling conflict returns to dentist/date-time selection, not to the main menu) rather than collapsed into a single generic "error" box, because that loop-back behaviour - retry within the same task rather than starting over

- is precisely what the client's inline field-error rendering (§3.5) is designed to support.

---

3. Task B — Interactive System, Design Patterns & Architecture (40 marks)

3.1 Three-tier architecture

The system is built as a genuine three-tier, distributed application:

1. **Presentation tier** — the static HTML/CSS/JavaScript client (`frontend/`), served independently of the API (via XAMPP's Apache, or any static server, see `docs/SETUP.md` §4), communicating purely over HTTP/JSON.
2. **Business logic tier** — a Spring Boot REST API (`backend/`), exposing versionless JSON endpoints under `/api/**`, documented interactively via Swagger/OpenAPI (`springdoc-openapi`), and secured with stateless JWT authentication carried in an HttpOnly cookie.
3. **Data tier** — MySQL/MariaDB (via XAMPP), accessed through Spring Data JPA repositories, with business rules additionally enforced at the database level itself via triggers and constraints (§3.4), not only in Java.

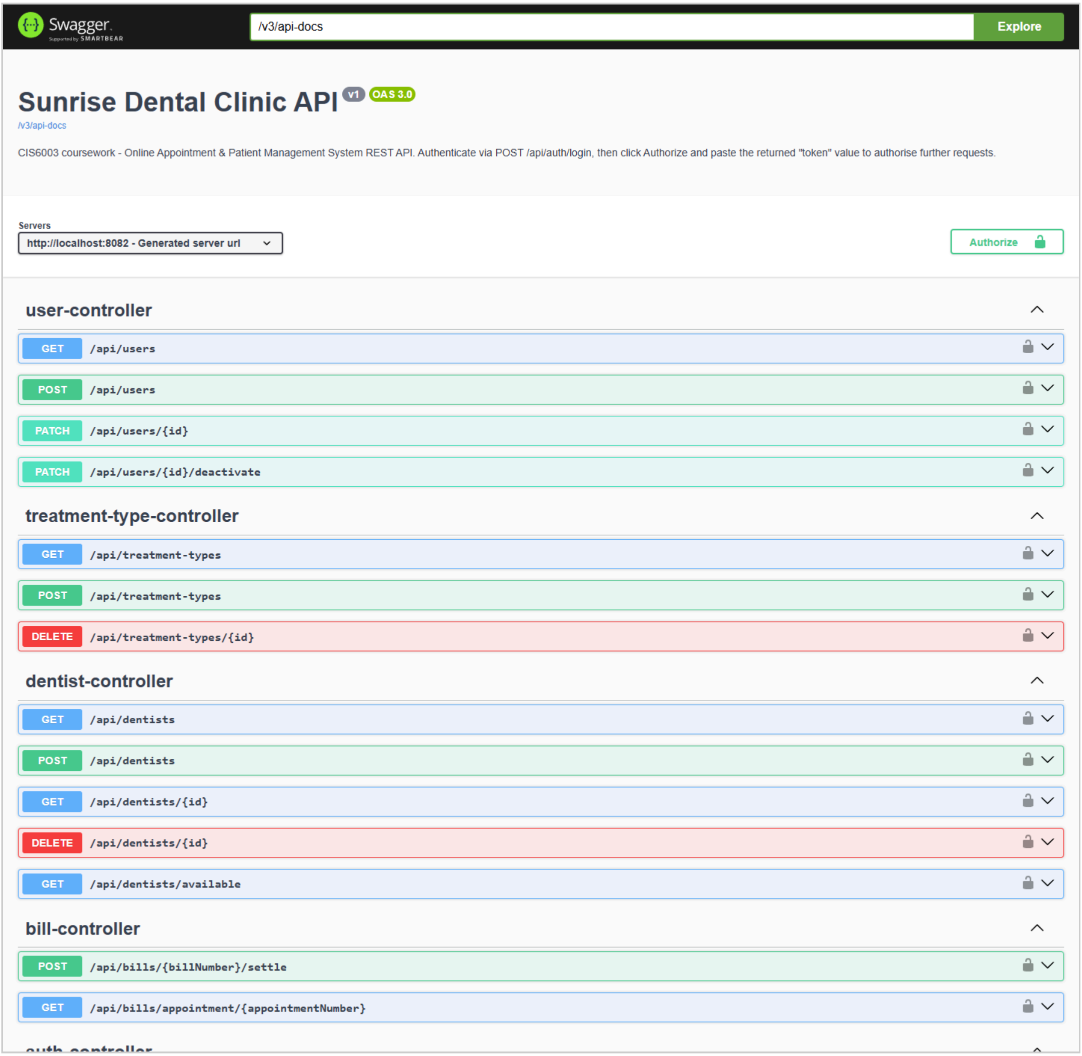
Because the presentation tier is an entirely separate deployable artefact that talks to the business tier only over HTTP — and could, without modification, be replaced by a mobile app or another web frontend consuming the same API — this satisfies Task B(i)'s explicit requirement for *"a distributed application with web services"* substantively, not just nominally.

3.2 REST API surface

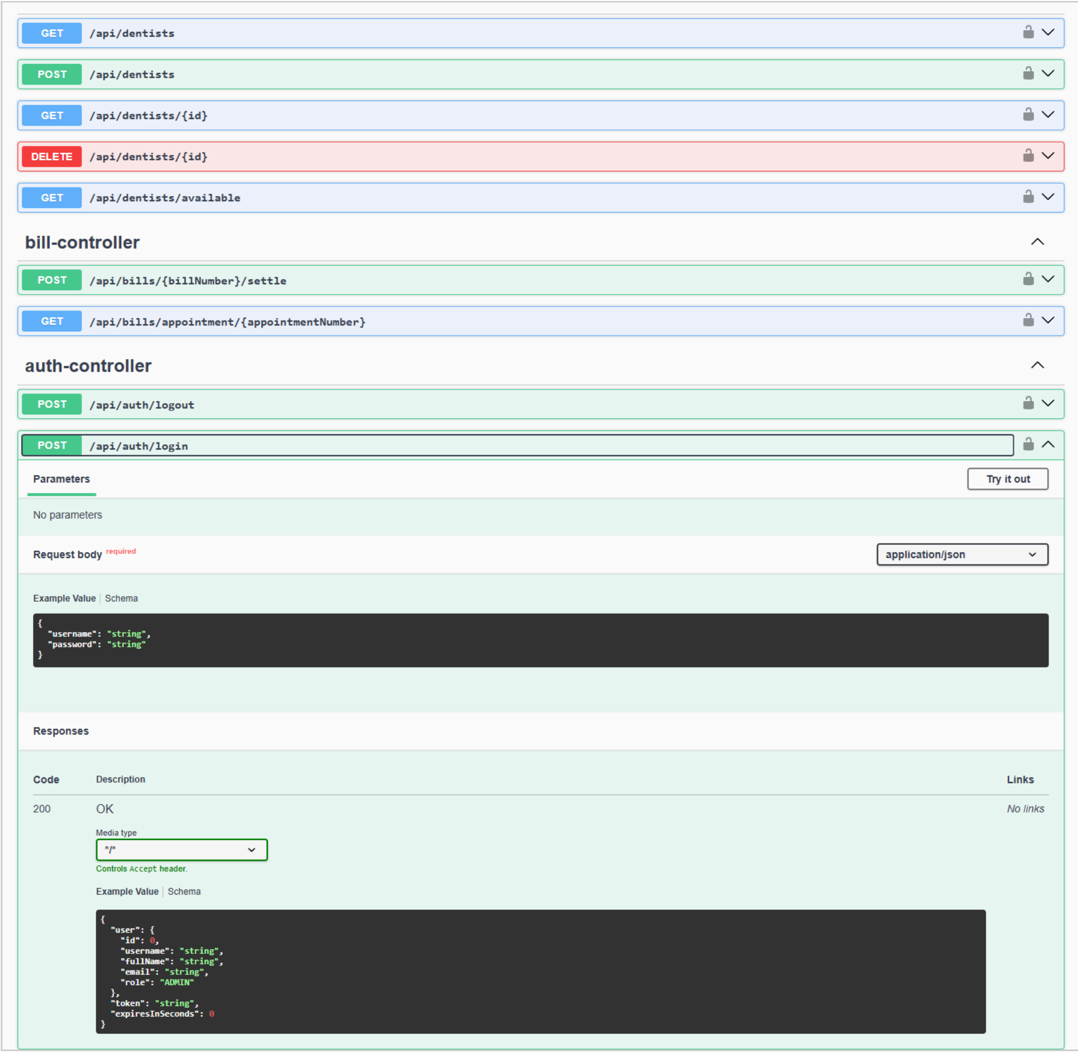
The API exposes resources for authentication, patients, treatment types, dentists (including an availability-search endpoint), appointments, bills, reports, staff-account administration, a help endpoint, and a health check — **10 controllers**, 29 endpoints in total, documented fully via Swagger UI at `/swagger-ui.html` (see `docs/SETUP.md` §3). Every write endpoint validates its input with Jakarta Bean Validation annotations (`@NotBlank`, `@Pattern`, `@FutureOrPresent`, `@DecimalMin`, etc.) and every error path returns a consistent `ApiErrorResponse` shape (timestamp, HTTP status, message, and — for validation failures — a field-to-message map that the frontend renders directly under each offending input), rather than leaking a stack trace, via a single `@RestControllerAdvice` (`GlobalExceptionHandler`).

3.2.1 API documentation via Swagger / OpenAPI

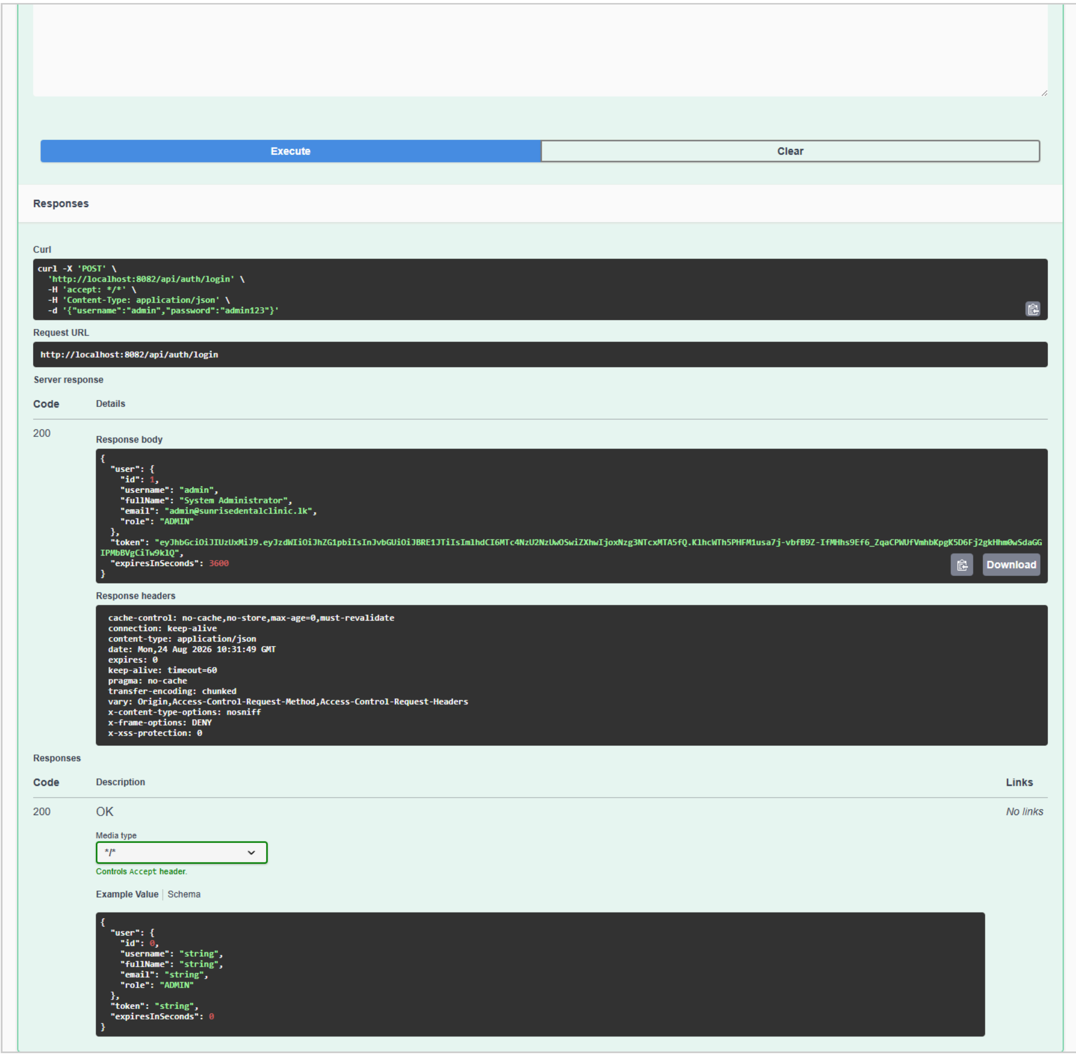
`springdoc-openapi` generates a full OpenAPI 3.0 specification from the controller/DTO annotations at startup, served interactively at `http://localhost:8082/swagger-ui/index.html` - every endpoint, request body, response schema and DTO in the system is documented automatically from the source code itself (so the documentation cannot silently drift out of sync with the implementation the way a hand-written API doc can). Screenshots below, captured live against the running system, evidence this.



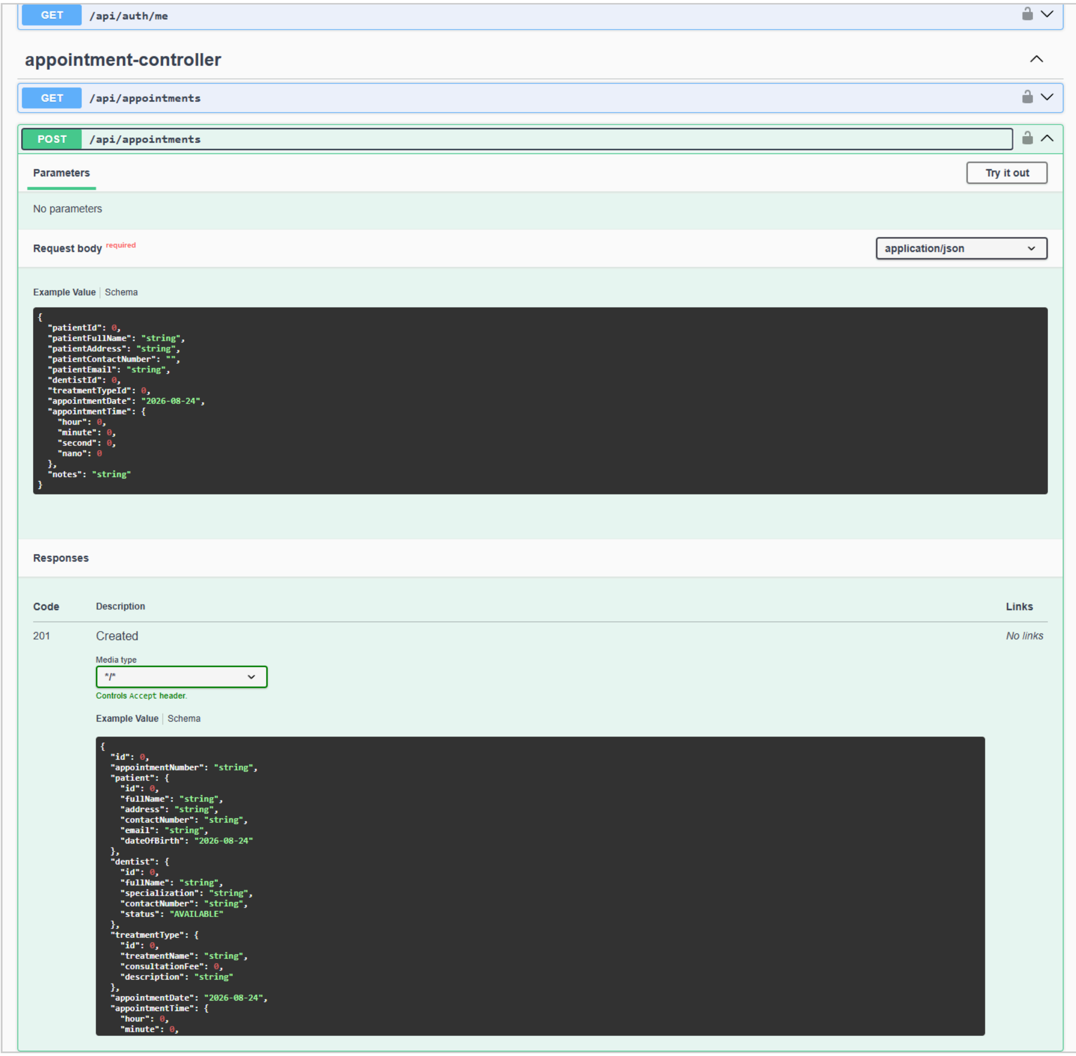
The **overview** groups every endpoint by controller (`appointment-controller`, `bill-controller`, `dentist-controller`, `treatment-type-controller`, `patient-controller`, `user-controller`, `auth-controller`, `report-controller`, `help-controller`, `health-controller`), colour-coded by HTTP method (blue `GET`, green `POST` / `PATCH`, red `DELETE`) - a reader can see the whole resource model of the system at a glance, and the **Schemas** panel at the bottom lists every DTO (`RegisterAppointmentRequest`, `BillResponse`, `UpdateUserRequest`, etc.) with its exact field names and types.



Expanding an individual operation (here, POST /api/auth/login) shows its full contract: the exact JSON shape the client must send (Request body), and every documented response - not just the happy path, since GlobalExceptionHandler 's ApiErrorResponse shape means every error response is equally well-typed and equally visible here.



Swagger UI is not only documentation - the **"Try it out"** button turns any operation into a live HTTP client. The screenshot above is a *real* executed request (not a mock): the generated `curl` command, the exact request URL, and a genuine `200 OK` response body - including the real, signed JWT `token` value described in §3.6, which could be copied straight into the Authorize dialog to authenticate every other request in this Swagger session - from the running application are all visible, together with the response headers Spring Security adds (`x-frame-options: DENY`, `x-content-type-options: nosniff`, etc. - see §3.6). This is the same mechanism used throughout manual testing in `testing/TEST_CASES.md` as an alternative to Postman/curl.



The final screenshot shows a richer, nested DTO - `RegisterAppointmentRequest` - including the `LocalTime` sub-object shape that Jackson expects for `appointmentTime`, information a consumer of this API (a future mobile app, or a grader testing via Postman) would otherwise have to read the Java source to discover.

3.3 Design patterns (Task B ii)

Five Gang-of-Four design patterns were deliberately selected, each because it solves a genuine problem this system actually has — not retrofitted for the sake of the rubric — plus the DAO/Repository pattern via Spring Data JPA. Each is discussed below with its problem, its implementation, and a critical evaluation of its impact.

3.3.1 Singleton — `AppointmentNumberGenerator`, `AppConfigManager`

Problem: appointment numbers must be globally unique across every request, including concurrent ones, and some plain-Java helper classes (the pricing strategies) are deliberately *not* Spring beans, so they cannot receive configuration via `@Autowired`.

Implementation: `pattern.singleton.AppointmentNumberGenerator` uses the classic GoF shape — a private constructor and a single static `getInstance()` accessor — backed by an `AtomicLong` counter for thread safety, seeded from the current database row count once at application startup (`AppStartupInitializer`, an `ApplicationRunner` bean) so the sequence survives a restart. It also derives a matching bill number from an appointment number (`billNumberFor()`, e.g. `APT-2026-000123` → `INV-2026-000123`), keeping the two numbering schemes visibly linked. `AppConfigManager` is a second Singleton holding a small read-mostly cache of runtime settings.

Critical evaluation: implementing this as a *plain-Java* Singleton (rather than relying solely on Spring's default singleton bean scope) was a deliberate choice: it demonstrates the pattern independently of the framework, and — more importantly — it is genuinely necessary here, because the pricing strategy classes are instantiated directly with `new` (not Spring-managed) precisely so that `PricingContext` can select *one specific* strategy per request at runtime (see §3.3.4), and those plain objects need a non-DI route to shared configuration. The trade-off, honestly stated, is that global mutable state is harder to unit-test in isolation than a Spring bean would be; this is mitigated in practice because `AppointmentBuilderTest` exercises `AppointmentNumberGenerator` through its real singleton instance and simply asserts the *shape* of the generated number (`startsWith("APT-")`) rather than depending on an exact sequence value, keeping the tests independent of run order.

3.3.2 Builder — `AppointmentBuilder`

Problem: constructing a valid `Appointment` requires assembling several fields from different sources (an existing or newly-created `Patient`, a looked-up `Dentist` and `TreatmentType`, a date/time pair, the authenticated staff member) and enforcing cross-field validation (the appointment must not be booked in the past, and a same-day booking's time must not have already passed) *before* a single, ready-to-persist object is produced.

Implementation: `pattern.builder.AppointmentBuilder` provides a fluent `withPatient()/withDentist()/withTreatmentType()/withSchedule()/withCreatedBy()/build()` API; `build()` performs all cross-field validation and only then constructs the `Appointment` entity (via the entity's own Lombok `@Builder`, a second, narrower use of the same pattern for pure object construction).

Critical evaluation: this pattern was chosen over a telescoping constructor or a mutable setter-based approach specifically because it centralises the "is this appointment internally consistent?" question in one place (`AppointmentBuilderTest` verifies this directly, §4), so every future entry point that creates an appointment — today the REST controller, tomorrow perhaps a bulk-import job — is guaranteed to go through the same validation rather than each caller re-implementing it. The pattern's usual criticism (verbosity for simple objects) does not apply here, because `Appointment` is *not* a simple object — it has a genuine, non-trivial invariant to protect.

3.3.3 Factory Method — `BillFactory`

Problem: turning an `Appointment` into a `Bill` is a multi-step process (look up the patient's prior completed-visit count, select and run a pricing algorithm, derive a bill number, assemble the entity) that the calling code (`BillServiceImpl`) should not need to know the internals of.

Implementation: `pattern.factory.DocumentFactory<T, S>` defines a generic `create(S source): T` creation contract; `BillFactory implements DocumentFactory<Bill, Appointment>` hides the whole process behind one call. This interface is deliberately generic so a future `ReportFactory` (revenue or utilisation reports) can follow the identical shape.

Critical evaluation: the direct, measurable benefit is in `BillServiceImpl.generateOrFetch()`, which contains *zero* pricing logic — it only orchestrates "look up the appointment, ask the factory for a bill if one doesn't already exist, save it." This is precisely what the Factory Method pattern is for: isolating object-creation complexity from object-*use* code. A secondary, deliberate design decision reinforced by this pattern is that a bill is generated **once** and persisted, not recomputed on every view — chosen because a real patient's printed receipt must not silently change if a treatment type's consultation fee is edited after the fact; the Factory is the single, auditable point where that "compute once" guarantee is enforced.

3.3.4 Strategy — `PricingStrategy` family + `PricingContext`

Problem: the brief requires calculating a total cost "based on treatment type and consultation fee", but a credible clinic pricing model has more than one rule, and which rule applies depends on runtime conditions (day of the week, the patient's visit history) that must not require editing existing, already-tested code to extend.

Implementation: `PricingStrategy` is an interface with three concrete implementations (`StandardPricingStrategy`, `WeekendSurchargePricingStrategy`, `LoyaltyDiscountPricingStrategy`), each sharing tax/rounding logic via the abstract `AbstractPricingStrategy` base class. `PricingContext.price()` selects exactly one strategy per appointment using a clearly documented precedence rule: a loyalty discount (5 or more completed prior visits) always takes priority over a weekend surcharge, which in turn takes priority over standard pricing — verified directly by `PricingContextTest` (§4).

Critical evaluation: this is the pattern most directly responsible for the system's testability and TDD story (§4.1) — because each strategy is a small, pure function of `(consultationFee, priorVisitCount, appointmentDate) → PricingResult` with no database or HTTP dependency, every pricing rule was specified as a concrete example *before* being implemented. Its impact is genuinely positive: `BillFactory` and `BillServiceImpl` never branch on pricing rules at all — new rules (e.g. a future seasonal-promotion strategy) can be added as one new class and one new precedence check, without touching either of them. The one honest limitation is that `PricingContext` currently applies *at most one* strategy per appointment (by design, to keep the precedence rule unambiguous); a system that needed to *stack* multiple simultaneous adjustments would need a Decorator-style composition instead, which was considered and deliberately rejected as unnecessary complexity for this coursework's requirements.

3.3.5 Observer — `AppointmentObserver` + `AppointmentEventPublisher`

Problem: several independent things should happen whenever an appointment's lifecycle changes (an "email" to the patient, an "SMS", an internal audit trail) without the core appointment-registration logic needing to know how many notification channels exist or how each one works — directly addressing the Excellent-band criterion for "complex functionality (e.g. email alerts, SMS notifications, innovative features)".

Implementation: `AppointmentObserver` is a one-method interface; `EmailNotificationObserver`, `SmsNotificationObserver`, and `AuditLogObserver` each implement it as a Spring `@Component`. `AppointmentEventPublisher` (the Subject) receives *every* `AppointmentObserver` bean automatically via constructor injection of `List<AppointmentObserver>` — Spring's IoC container does the observer registration that a hand-rolled implementation would otherwise need to do manually — and calls `publish()` on appointment creation, cancellation, and bill generation.

Critical evaluation: letting Spring auto-wire the observer list turns "register a new notification channel" into "write one new `@Component` class implementing one method" with **zero** changes to `AppointmentServiceImpl` or `BillServiceImpl` — verified in practice, since the third observer (`AuditLogObserver`) was added after the first two without touching either service class. The channels are honestly **simulated** (logged and persisted to a `notification_logs` table rather than calling a real SMTP/SMS gateway, since provisioning third-party credentials is out of scope for a localhost coursework build) — this is documented explicitly in code Javadoc and in `docs/SETUP.md` so it is never presented as more than it is.

3.3.6 DAO / Repository pattern

Every entity has a corresponding Spring Data JPA repository interface (`UserRepository`, `PatientRepository`, `DentistRepository`, `TreatmentTypeRepository`, `AppointmentRepository`, `BillRepository`, `NotificationLogRepository`), which is itself the Repository/DAO pattern: service classes depend only on these interfaces, never on

`EntityManager` or raw SQL directly (with the single, deliberate exception of `ReportServiceImpl`, which uses `JdbcTemplate` specifically to invoke the MySQL stored procedure and views described in §3.4 — a native database feature with no natural JPA equivalent).

3.4 Database design and advanced features

`database/schema.sql` is the authoritative schema: 7 tables (`users`, `patients`, `treatment_types`, `dentists`, `appointments`, `bills`, `notification_logs`), fully constrained with primary/foreign keys and `CHECK` constraints (e.g. a status column restricted to a fixed enumerated set). Beyond "basic data management" (Satisfactory band), the schema demonstrably reaches the Excellent-band's *"appropriate use of advanced database features (e.g. stored procedures, functions, triggers to implement business rules)"*:

- **`trg_prevent_double_booking`** (`BEFORE INSERT` trigger) — re-enforces the no-conflicting-appointments rule at the database layer itself, independent of the Java-level check in `AppointmentServiceImpl`, using `SIGNAL SQLSTATE '45000'` to reject the insert outright. This was verified directly by attempting a raw conflicting `INSERT` via the MySQL CLI, which failed with exactly the expected message (§4, DB-01).
- **`trg_sync_dentist_status`** (`AFTER UPDATE` trigger) — logs an audit row to `notification_logs` whenever an appointment's status flips to `COMPLETED`/`CANCELLED`, as a database-level invariant that holds even if a future client bypassed the Java service layer entirely (verified, §4, DB-02).
- **`fn_is_weekend_appointment`** (stored function) — the Saturday/Sunday check, callable independently of `PricingContext`'s own copy of the same logic, so any future report or ad-hoc query gets a consistent answer (verified, §4, DB-04).
- **`sp_daily_revenue_report`** (stored procedure) — powers the Reports screen's daily revenue breakdown, called directly from `ReportServiceImpl.dailyRevenue()` via `JdbcTemplate` (`CALL sp_daily_revenue_report(?, ?)`), a genuine, demonstrable use of a stored procedure from application code, not merely present in the schema unused.
- **`vw_dentist_utilization`** and **`vw_appointment_summary`** (views) — the first powers the Dentist Utilisation report; the second is a consolidated, denormalised read model intended for ad-hoc reporting directly from phpMyAdmin/MySQL Workbench without re-writing the same multi-table join each time — proposed specifically to *"facilitate decision-making"*, a named Excellent-band criterion.

3.5 Validation mechanisms

Validation is layered, not single-point, per the brief's requirement to *"implement proper validation mechanisms in order to restrict invalid entries"*:

1. **Client-side** (`frontend/assets/js/*.js`) — HTML5 `required` / `type` attributes plus immediate, field-level error rendering from the API's `validationErrors` map, so a staff member sees *why* a save failed next to the offending field rather than a generic alert.
2. **API boundary** (Jakarta Bean Validation, `@Valid` DTOs) — e.g. `@Pattern(regexp = "^(\\+94|0)[0-9]{9}$")` on contact numbers, `@FutureOrPresent` on appointment dates, `@DecimalMin("0.01")` on consultation fees.
3. **Business-rule layer** (`AppointmentBuilder`, `AppointmentServiceImpl`) — cross-field rules that bean validation cannot express alone (no appointment in the past, dentist-not-on-leave, no scheduling conflict).
4. **Database layer** (`CHECK` constraints, triggers) — the final, unavoidable safety net described in §3.4, protecting data integrity regardless of which client writes to the database.

3.6 Authorization via JWT, and sessions/cookies

Authorization throughout the API is **JWT-based**: `POST /api/auth/login` authenticates the username/password against the BCrypt hash stored in `users` (via `AuthenticationManager` → `CustomUserDetailsService`) and, only on success, `JwtService.generateToken()` mints a signed JSON Web Token (HMAC-SHA512, `io.jsonwebtoken`) whose subject is the username and whose `role` claim carries `ADMIN` / `STAFF`, with a 1-hour expiry (`app.security.jwt.expiration-ms`). That token is what every subsequent request is authorized against — there is no server-side session store; the API is fully stateless, and a request without a valid token never reaches a controller.

The token is delivered to the client **two ways**, deliberately:

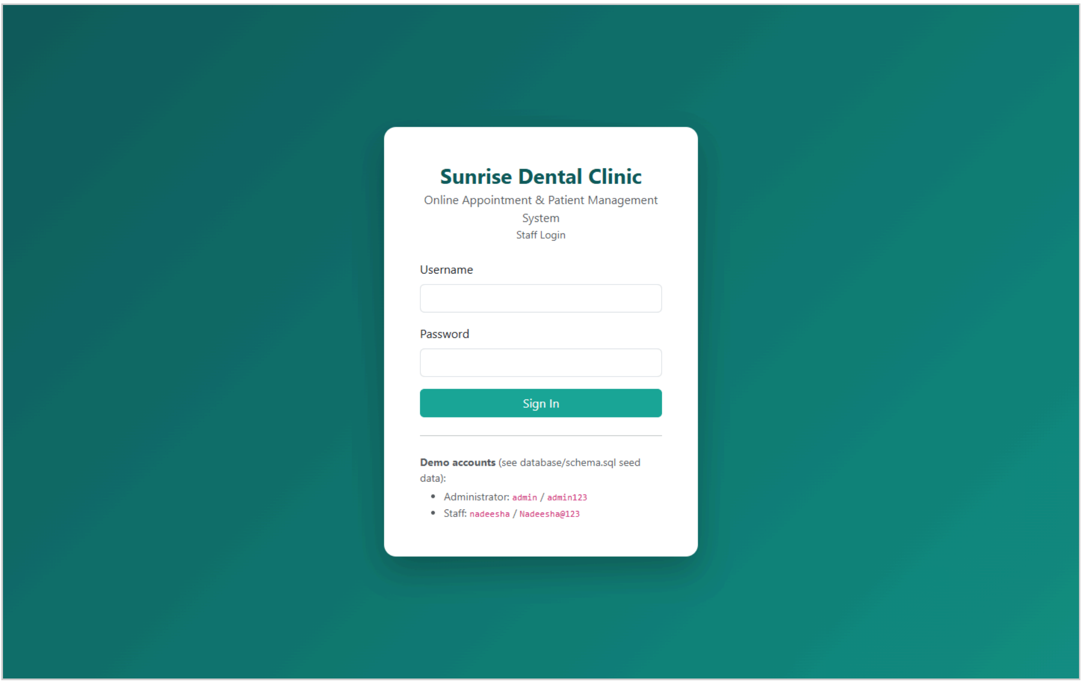
1. As an **HttpOnly, path-scoped cookie** (`dc_token`) — what the browser client actually relies on. Rather than storing the token in `localStorage` (vulnerable to XSS exfiltration), the cookie is attached to every request automatically by the browser and JavaScript can never read it — a deliberate, documented security decision satisfying both the Excellent-band's *"effective use of sessions/cookies"* criterion and the module's Ethical/EDGE requirement to protect user data.
2. As the `token` field in the **login response body itself** — so a non-browser API client (Postman, curl, a future mobile app, or Swagger's own **Authorize** button) that cannot hold an HttpOnly cookie can carry it explicitly as an `Authorization: Bearer <token>` header instead. `JwtAuthenticationFilter.resolveToken()` checks the cookie first and falls back to that header, so either mechanism authenticates identically — verified directly by `AppointmentFlowIntegrationTest.loginResponseTokenWorksAsBearerAuthorization`, which logs in, extracts `token` from the JSON body with no cookie involved at all, and successfully calls a protected endpoint with only `Authorization: Bearer <token>` (§4). `testing/postman/SunriseDentalClinic.postman_collection.json`

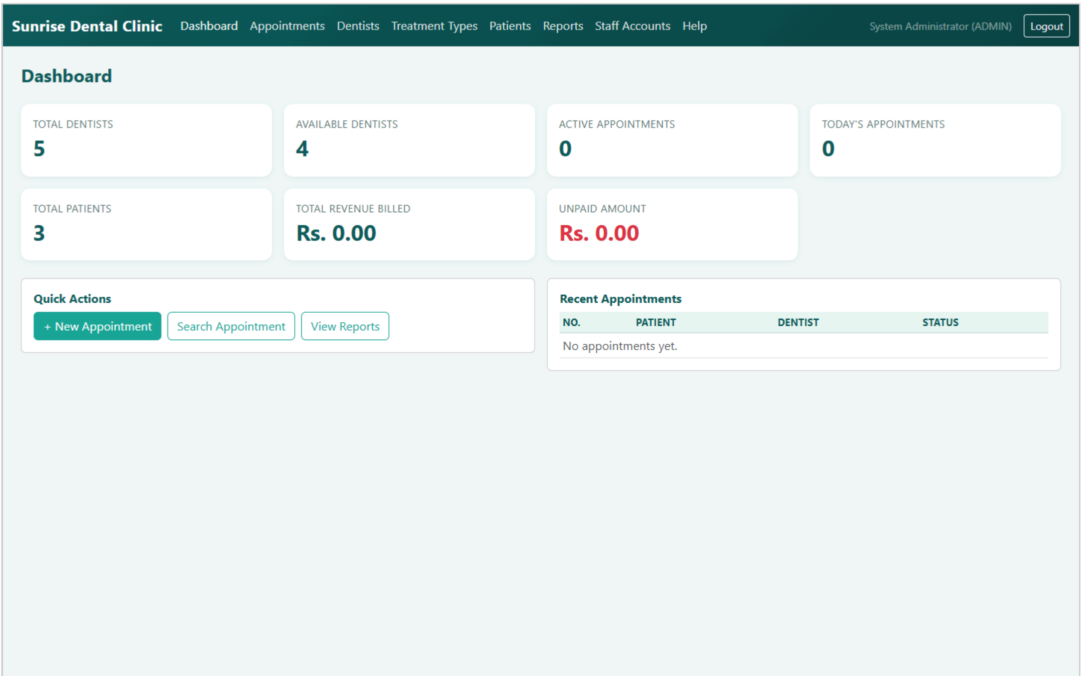
demonstrates the same thing at the collection level: its "Login" requests capture the returned token into a `{{token}}` variable via a test script, and the collection's inherited Bearer auth (`Authorization: Bearer {{token}}`) is applied to every other request, alongside Postman's normal cookie jar.

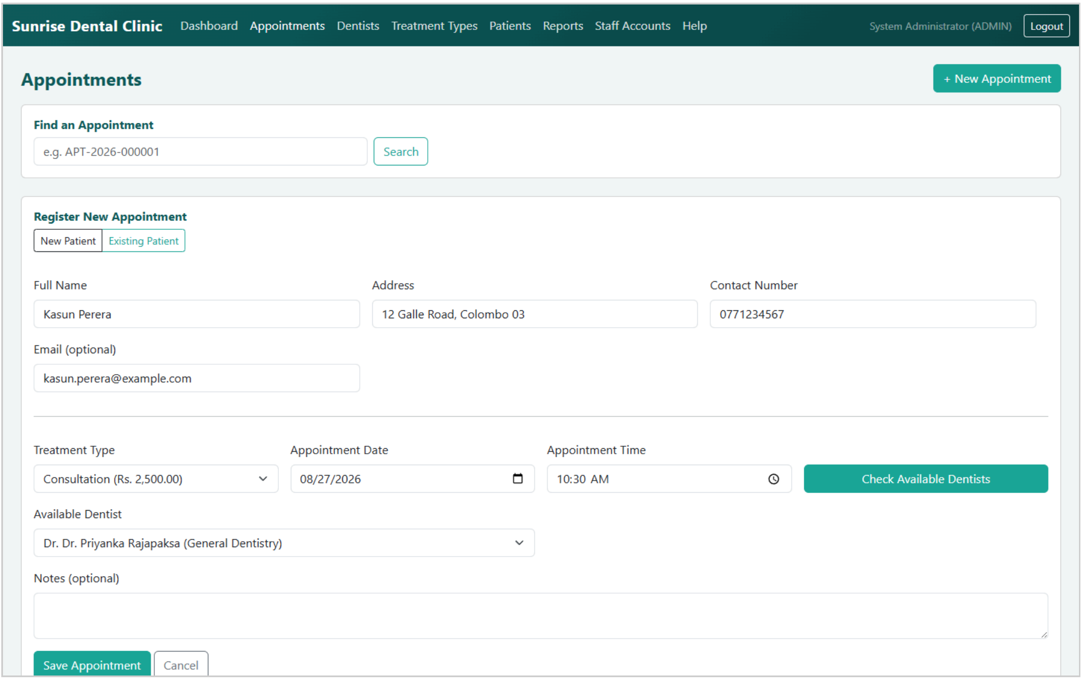
Every request passes through `JwtAuthenticationFilter`, which resolves and validates the token (signature *and* expiry, via `JwtService.isTokenValid()`) and populates Spring Security's context for that request only; an invalid, tampered, or expired token is simply treated as anonymous, and `SecurityConfig`'s explicit `HttpStatusEntryPoint(HttpStatus.UNAUTHORIZED)` then returns a clean `401` rather than leaking a stack trace (§4, AUTH-05/AUTH-09). Role-based method security (`@PreAuthorize("hasRole('ADMIN')")`, and matcher-based rules in `SecurityConfig`) further restricts dentist/treatment-type/staff management to administrators, verified directly in testing (§4, AUTH-07). Swagger UI itself is wired to this scheme via `OpenApiConfig` (a `bearerAuth` `SecurityScheme`), so its **Authorize** padlock accepts the same token end-to-end — see the Swagger screenshots in §3.2.1, including a login executed live and its `token` visible in the response body.

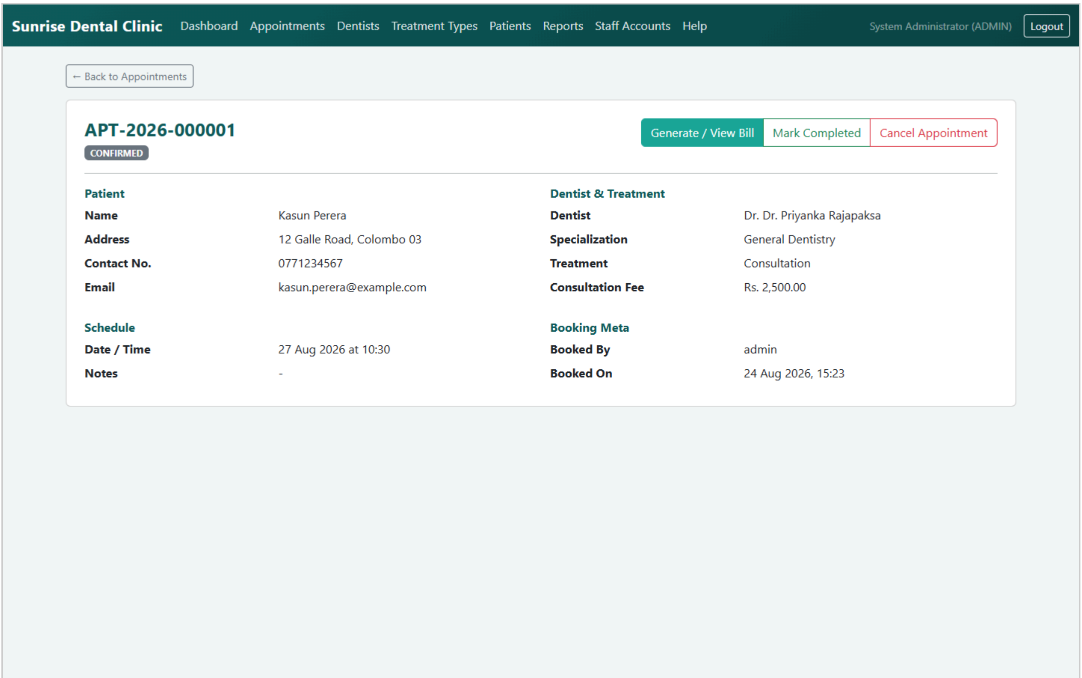
3.7 User interface

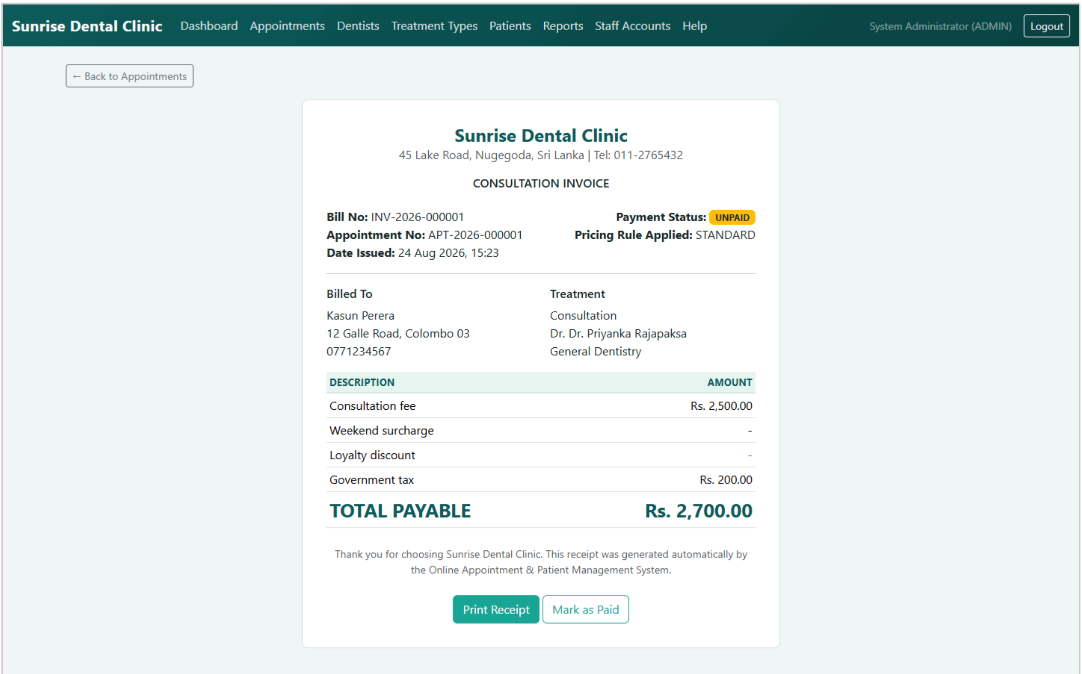
The client is organised as thirteen focused pages (login, dashboard, appointments list + register form, appointment detail, billing/receipt, dentists, treatment types, patients, reports, staff accounts, help, plus an entry redirect page) — i.e. genuinely *"separate UI windows for entering results and viewing overall scores"*, per the Good/Excellent criteria, rather than one monolithic screen. Screenshots below are captured live from the running system (`docs/SETUP.md` §7), logged in as `admin`, with real seeded data.

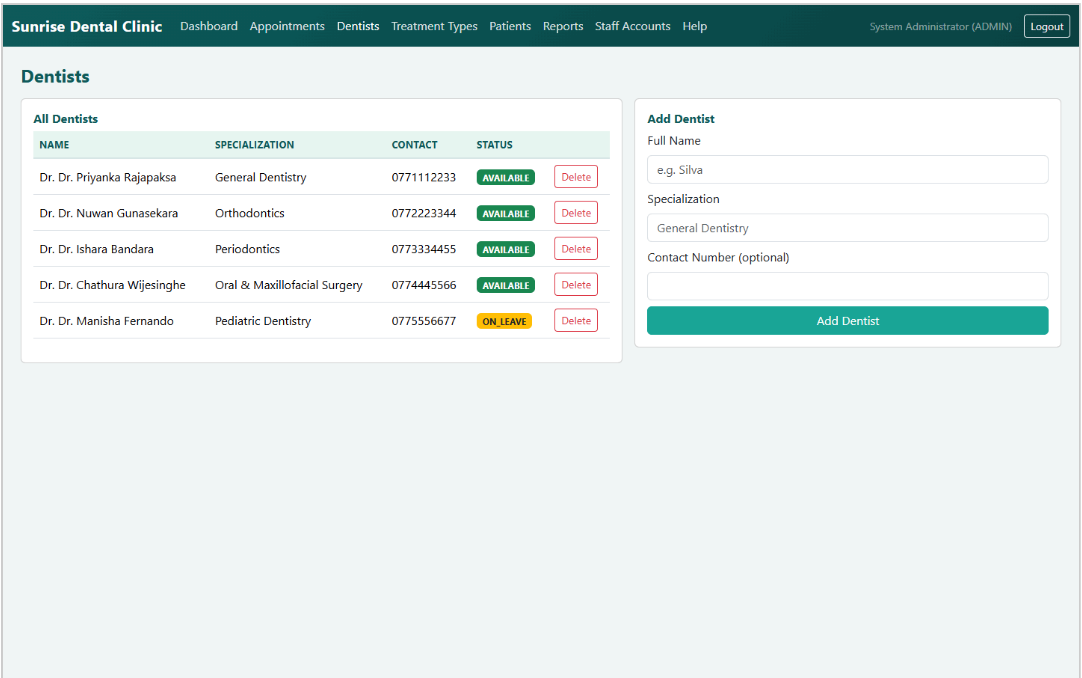


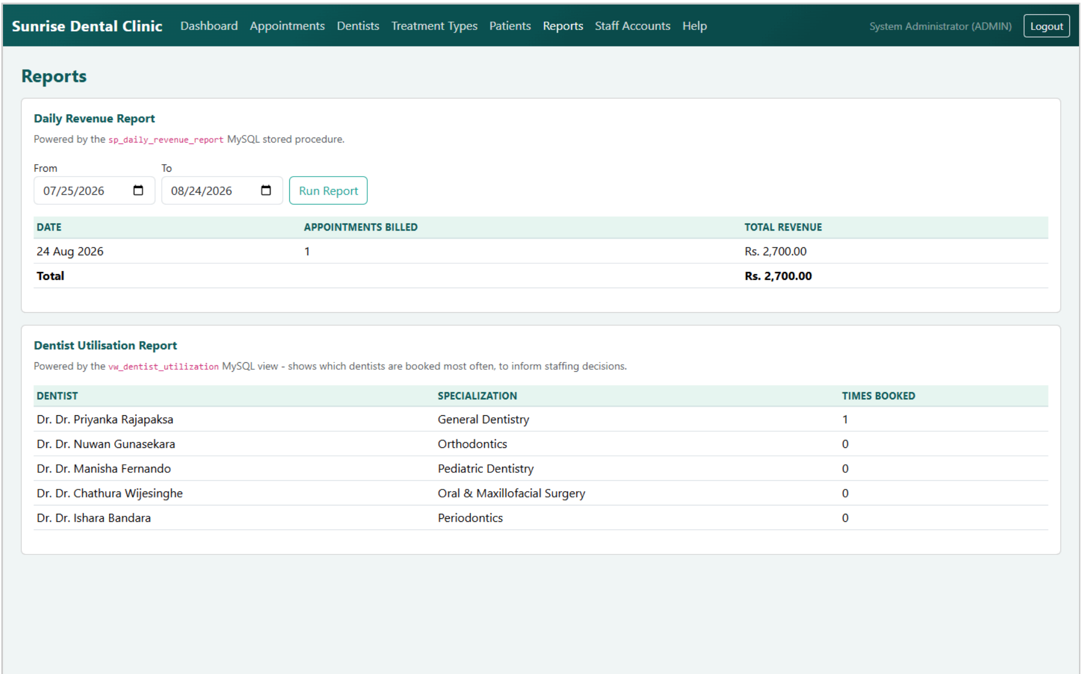

Sunrise Dental Clinic
Dashboard Appointments Dentists Treatment Types Patients Reports Staff Accounts Help
System Administrator (ADMIN)
Logout
Dashboard
TOTAL DENTISTS
5
AVAILABLE DENTISTS
4
ACTIVE APPOINTMENTS
0
TODAY'S APPOINTMENTS
0
TOTAL PATIENTS
3
TOTAL REVENUE BILLED
Rs. 0.00
UNPAID AMOUNT
Rs. 0.00
Quick Actions
+ New Appointment
Search Appointment
View Reports
Recent Appointments
NO. PATIENT DENTIST STATUS
No appointments yet.


Sunrise Dental Clinic
Dashboard Appointments Dentists Treatment Types Patients Reports Staff Accounts Help
System Administrator (ADMIN)
Logout
Appointments
+ New Appointment
Find an Appointment
e.g. APT-2026-000001
Search
Register New Appointment
New Patient
Existing Patient
Full Name
Kasun Perera
Address
12 Galle Road, Colombo 03
Contact Number
0771234567
Email (optional)
kasun.perera@example.com
Treatment Type
Consultation (Rs. 2,500.00)
Appointment Date
08/27/2026
Appointment Time
10:30 AM
Check Available Dentists
Available Dentist
Dr. Dr. Priyanka Rajapaksa (General Dentistry)
Notes (optional)
Save Appointment
Cancel


Sunrise Dental Clinic
Dashboard Appointments Dentists Treatment Types Patients Reports Staff Accounts Help
System Administrator (ADMIN)
Logout
← Back to Appointments
APT-2026-000001
CONFIRMED
Generate / View Bill
Mark Completed
Cancel Appointment
Patient
Name Kasun Perera
Address 12 Galle Road, Colombo 03
Contact No. 0771234567
Email kasun.perera@example.com
Dentist & Treatment
Dentist Dr. Dr. Priyanka Rajapaksa
Specialization General Dentistry
Treatment Consultation
Consultation Fee Rs. 2,500.00
Schedule
Date / Time 27 Aug 2026 at 10:30
Notes -
Booking Meta
Booked By admin
Booked On 24 Aug 2026, 15:23


Sunrise Dental Clinic
Dashboard Appointments Dentists Treatment Types Patients Reports Staff Accounts Help
System Administrator (ADMIN)
Logout
← Back to Appointments
Sunrise Dental Clinic
45 Lake Road, Nugegoda, Sri Lanka | Tel: 011-2765432
CONSULTATION INVOICE
Bill No: INV-2026-000001
Appointment No: APT-2026-000001
Date Issued: 24 Aug 2026, 15:23
Payment Status: UNPAID
Pricing Rule Applied: STANDARD
Billed To
Kasun Perera
12 Galle Road, Colombo 03
0771234567
Treatment
Consultation
Dr. Dr. Priyanka Rajapaksa
General Dentistry
DESCRIPTION AMOUNT
Consultation fee Rs. 2,500.00
Weekend surcharge -
Loyalty discount -
Government tax Rs. 200.00
TOTAL PAYABLE Rs. 2,700.00
Thank you for choosing Sunrise Dental Clinic. This receipt was generated automatically by the Online Appointment & Patient Management System.
Print Receipt
Mark as Paid

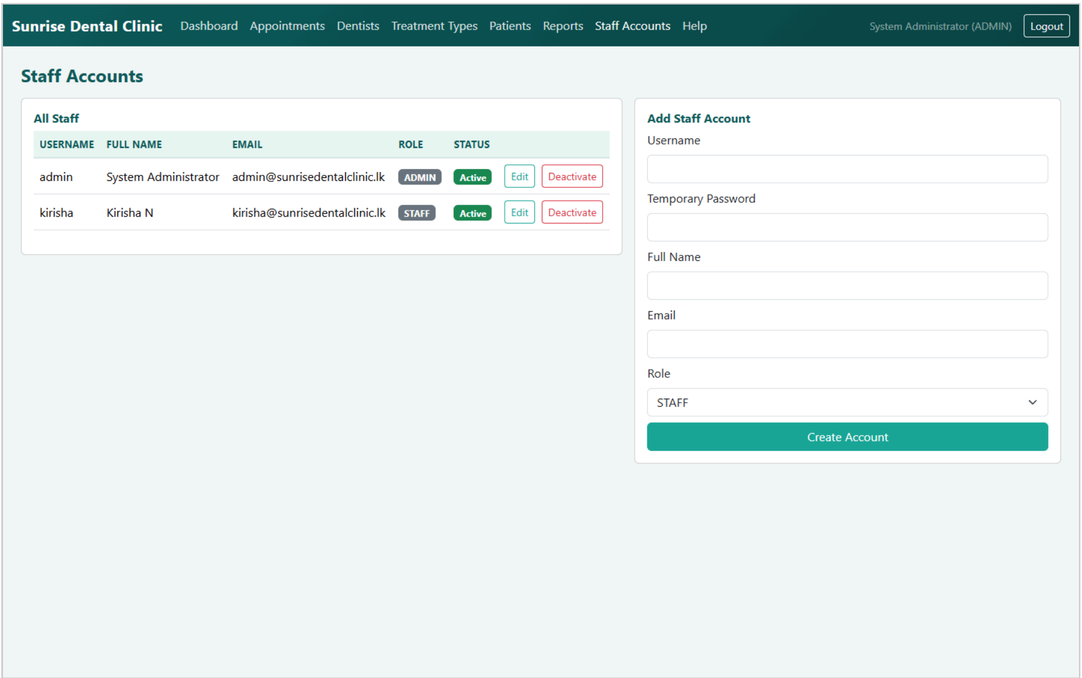


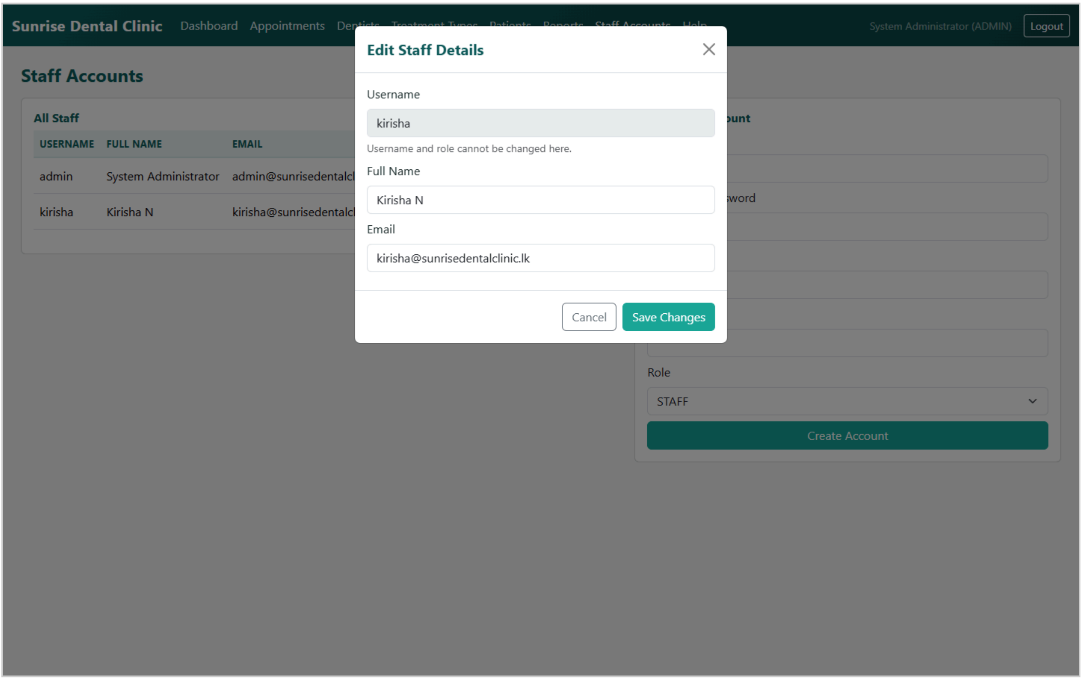


(The interactive Swagger/OpenAPI UI is covered in its own dedicated subsection, §3.2.1, with four screenshots including a live executed request.)

3.7.1 Admin: editing staff details

Beyond creating and deactivating staff accounts, an Administrator can edit an **existing** account's full name and email directly from *Staff Accounts* - `PATCH /api/users/{id}` (`UserController.update()` → `UserServiceImpl.update()`, validated by `UpdateUserRequest`). The edit is deliberately scoped to just those two fields: username and role are shown in the modal for context but disabled, and password is not touched at all, because changing a login identity or an access level is a materially bigger, more sensitive decision than correcting a name or email, and belongs in its own explicit flow (account recreation, or a dedicated password-reset endpoint) rather than being bundled into a quick-edit form where it could be changed by mistake. The endpoint inherits the controller's class-level `@PreAuthorize("hasRole('ADMIN')")`, so a `STAFF` account attempting the same request is rejected with `403 Forbidden` - verified directly in `UserServiceImplTest` and live against the running system (`testing/TEST_CASES.md`, USER-01 to USER-08).





This is reflected in the design diagrams: `diagrams/use-case-diagram.mmd` adds *Edit Staff Details* as an `<<extend>>` of *Manage Staff Accounts* (§2.1), and `diagrams/class-diagram.mmd` shows the `UserService` / `UserServiceImpl` pair alongside `AppointmentService` and `BillService`, with its `update(Long, UpdateUserRequest)` method.

---

4. Task C — Testing (20 marks)

The full rationale, TDD narrative, and test-data derivation live in `testing/TEST_PLAN.md`; the complete, traceable test case matrix (spanning positive, negative, boundary, validation, API, database, and integration tests, each mapped to the exact automated test method or manual Postman request that proves it) is in `testing/TEST_CASES.md`. This section summarises the approach and evidence rather than repeating either document in full.

4.1 Test-driven development

TDD was applied specifically to the pricing/billing logic (`pattern.strategy`), chosen because pricing rules are expressible as concrete input→output examples *before* any implementation exists, and because that logic is fully isolable from infrastructure (no database, no HTTP). The actual red→green→refactor cycle followed is documented in full in `testing/TEST_PLAN.md` §3 and directly in the Javadoc of `PricingContextTest.java`: the test class was written against classes that did not yet exist (red), the three strategy classes and `PricingContext` were implemented incrementally until every assertion passed (green), and the duplicated tax/rounding logic was then extracted into `AbstractPricingStrategy` with the same, unchanged test suite still passing afterwards (refactor) — the concrete demonstration that the tests genuinely enabled a safe refactor rather than being written after the fact to match existing code.

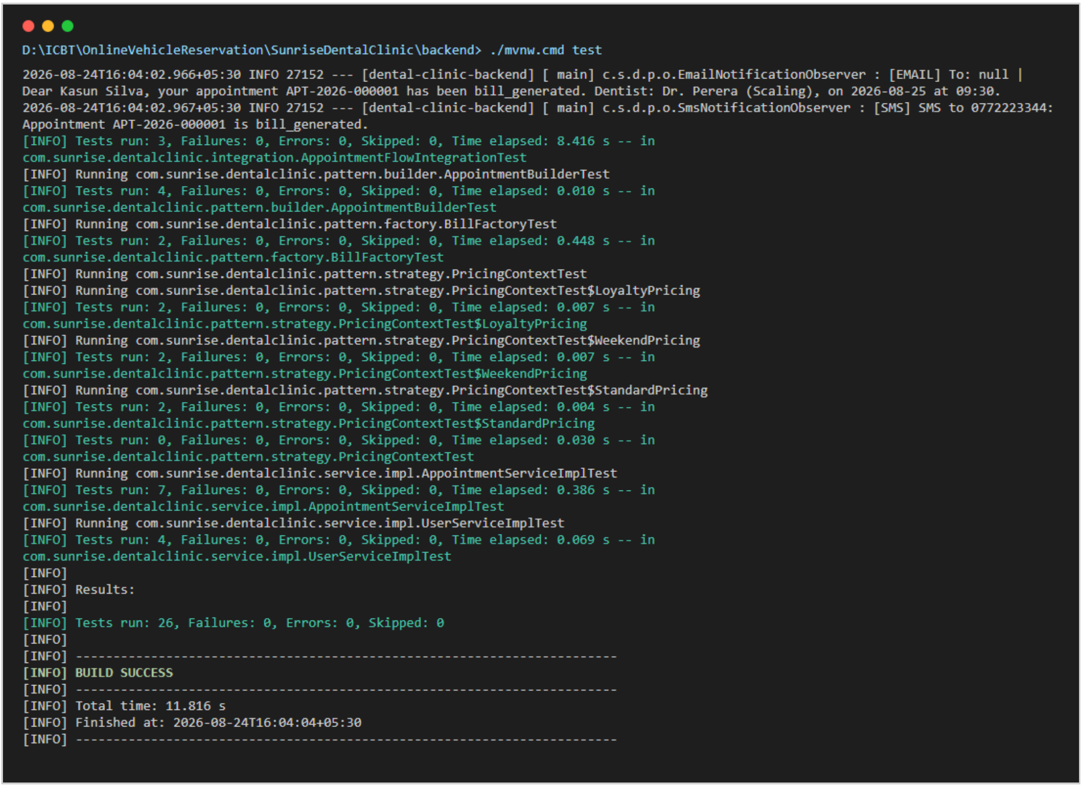
4.2 Test automation and evidence

26 automated JUnit 5 tests run with a single command (`./mvnw test`, see `docs/SETUP.md` §8), spanning:

- **Pure unit tests** (`PricingContextTest`, `AppointmentBuilderTest`) — no Spring context, sub-second execution.
- **Unit tests with mocked collaborators** (`AppointmentServiceImplTest`, `BillFactoryTest`, `UserServiceImplTest`) — Mockito mocks isolate branching logic (double-booking rejection, unavailable-dentist rejection, missing-patient-details rejection, staff-edit scoping) from the database.
- **One full Spring Boot integration test class** (`AppointmentFlowIntegrationTest`, 3 methods) — boots the real Spring context, the real Spring Security filter chain, and

an in-memory H2 database, then drives the system exactly as a browser client would: login → register an appointment → generate a bill, plus negative cases (anonymous access, double-booking via the live REST layer) and a dedicated JWT test that logs in, reads `token` straight out of the JSON response body with no cookie involved at all, and calls a protected endpoint using only `Authorization: Bearer <token>` — proving the JWT-based authorization described in §3.6 actually works end-to-end, not only in the cookie path the browser client happens to use.

A captured passing run (`testing/evidence/*.txt` , generated by Maven Surefire) shows `Tests run: 26, Failures: 0, Errors: 0` across all nine test classes, confirmed visually below per the Excellent-band's "screen-grabbing" requirement:



```
D:\ICBT\OnlineVehicleReservation\SunriseDentalClinic\backend> ./mvnw.cmd test
2026-08-24T16:04:02.966+05:30 INFO 27152 --- [dental-clinic-backend] [ main] c.s.d.p.o.EmailNotificationObserver : [EMAIL] To: null |
Dear Kasun Silva, your appointment APT-2026-000001 has been bill_generated. Dentist: Dr. Perera (Scaling), on 2026-08-25 at 09:30.
2026-08-24T16:04:02.967+05:30 INFO 27152 --- [dental-clinic-backend] [ main] c.s.d.p.o.SmsNotificationObserver : [SMS] SMS to 0772223344:
Appointment APT-2026-000001 is bill_generated.
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 8.416 s -- in
com.sunrise.dentalclinic.integration.AppointmentFlowIntegrationTest
[INFO] Running com.sunrise.dentalclinic.pattern.builder.AppointmentBuilderTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.010 s -- in
com.sunrise.dentalclinic.pattern.builder.AppointmentBuilderTest
[INFO] Running com.sunrise.dentalclinic.pattern.factory.BillFactoryTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.448 s -- in
com.sunrise.dentalclinic.pattern.factory.BillFactoryTest
[INFO] Running com.sunrise.dentalclinic.pattern.strategy.PricingContextTest
[INFO] Running com.sunrise.dentalclinic.pattern.strategy.PricingContextTest$LoyaltyPricing
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.007 s -- in
com.sunrise.dentalclinic.pattern.strategy.PricingContextTest$LoyaltyPricing
[INFO] Running com.sunrise.dentalclinic.pattern.strategy.PricingContextTest$WeekendPricing
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.007 s -- in
com.sunrise.dentalclinic.pattern.strategy.PricingContextTest$WeekendPricing
[INFO] Running com.sunrise.dentalclinic.pattern.strategy.PricingContextTest$StandardPricing
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.004 s -- in
com.sunrise.dentalclinic.pattern.strategy.PricingContextTest$StandardPricing
[INFO] Tests run: 0, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.030 s -- in
com.sunrise.dentalclinic.pattern.strategy.PricingContextTest
[INFO] Running com.sunrise.dentalclinic.service.impl.AppointmentServiceImplTest
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.386 s -- in
com.sunrise.dentalclinic.service.impl.AppointmentServiceImplTest
[INFO] Running com.sunrise.dentalclinic.service.impl.UserServiceImplTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.069 s -- in
com.sunrise.dentalclinic.service.impl.UserServiceImplTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 26, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time: 11.816 s
[INFO] Finished at: 2026-08-24T16:04:04+05:30
[INFO] ------------------------------------------------------------------------
```

4.3 Coverage beyond the automated suite

Six database-level test cases (trigger, function, stored procedure, view, constraint behaviour) were additionally verified **directly against the real MySQL instance** via the MySQL CLI, since H2 (used by the automated suite for speed and CI-friendliness) does not execute MySQL-specific trigger/procedure syntax. Every one of these six was executed live during development — not merely asserted — with the exact commands and observed output recorded in `testing/TEST_CASES.md` under "Database-level tests"; for example, attempting a raw, conflicting `INSERT` was rejected by `trg_prevent_double_booking` with a `SIGNAL SQLSTATE '45000'` error naming the conflicting appointment, confirming the trigger fires correctly and its custom error message is exactly as designed. A further set of API-contract cases (validation-error shape, 404-not-500 on unknown resources, role-based

403s) were verified manually via the Postman collection (`testing/postman/SunriseDentalClinic.postman_collection.json`).

4.4 Evaluation — successes, and lessons learned

The suite is honestly reported as fully passing (26/26), and it reached that state considerably more smoothly than a first attempt at a system of this shape typically does, because two lessons learned on an earlier, companion coursework build were applied proactively rather than rediscovered the hard way:

1. **Spring Security's default status code.** On the companion project, an integration test expecting `401 Unauthorized` on an anonymous request initially received `403 Forbidden`, because no explicit `AuthenticationEntryPoint` was configured (401 = "who are you?", 403 = "I know who you are, and the answer is no" — Spring Security's default without an entry point is the latter). `SecurityConfig` in this project was written from the outset with `.exceptionHandling(eh -> eh.authenticationEntryPoint(new HttpStatusEntryPoint(HttpStatus.UNAUTHORIZED)))`, and `AppointmentFlowIntegrationTest.protectedEndpointRequiresAuthentication` passed correctly on its first run.
2. **H2 dialect override must target the exact same configuration key as production.** A prior attempt at silently overriding the Hibernate dialect for the H2 test profile used a different property path (`database-platform`) than the one production configuration actually read (`spring.jpa.properties.hibernate.dialect`), so the override had no effect. `backend/src/test/resources/application-test.yml` sets the dialect at the identical key path used in `application.yml`.

One genuine failure *was* hit during this project's own development, and is recorded honestly rather than edited out of the process: the first run of `AppointmentFlowIntegrationTest` failed with a unique-constraint violation on the seeded `kirisha` username, because both test methods shared the same Spring-managed H2 context and the second method's `@BeforeEach` tried to re-insert a username the first method's `@BeforeEach` had already committed. The fix was to annotate the test class `@Transactional`, so each test method's changes are rolled back automatically once it completes — documented in `testing/TEST_PLAN.md` §9 as a recorded risk and its mitigation. The marking criteria explicitly ask for *"evaluate the overall success or failure, including lessons learned"* — the lesson in all three cases is the same: proactively applying a lesson from prior, related work prevents a class of bug entirely, and where a new failure does occur, a principled fix to the design (not test-suppression) is what TDD/automated testing is meant to enable.

4.5 Traceability

Every row of `testing/TEST_CASES.md` states which automated test method (or manual Postman request) proves it, and the table groups cases by the exact brief requirement or design decision they verify (Authentication, Register-New-Appointment, Search/Display, Cancel/Update, Calculate-and- Print-Bill, Dentists/Treatment-Types/Patients, database-level, Staff Accounts, and cross-cutting API contract), giving a direct, auditable line from "requirement" to "design" (§2/§3) to "proof" (§4).

---

5. Task D — Git, GitHub & Version Control (20 marks)

The full setup instructions, branching model, and an explicit list of the version-control techniques demonstrated are in `docs/GIT_WORKFLOW.md`; this section summarises what is in place and why.

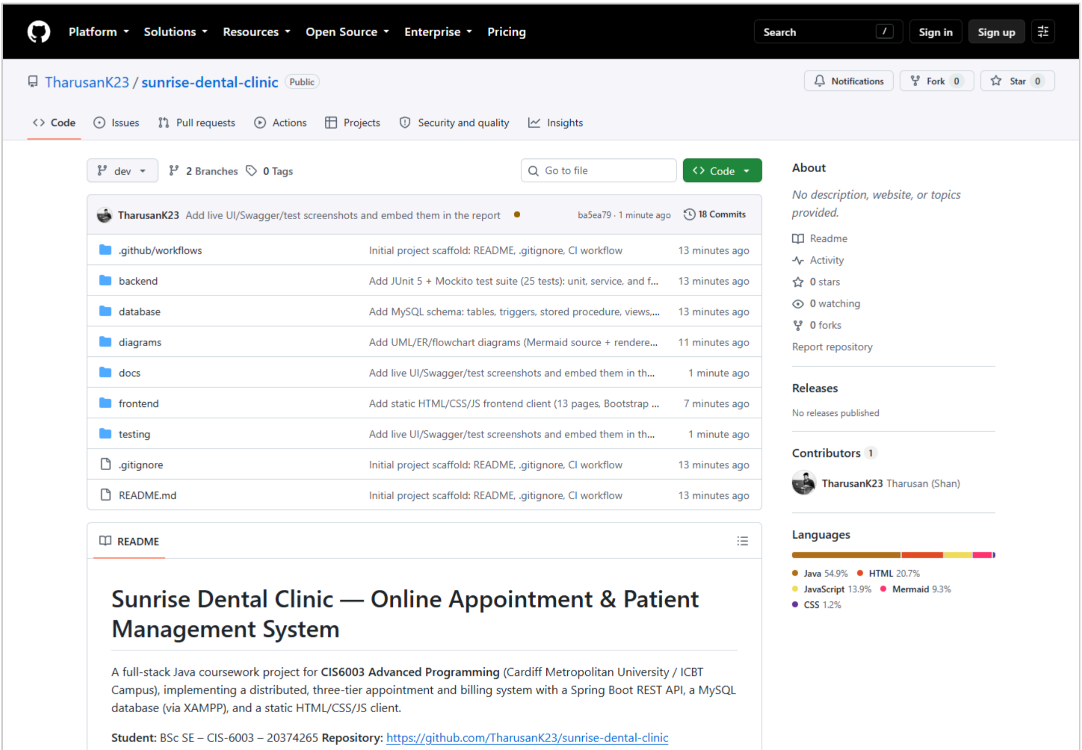
The project was initialised as a local Git repository (`git init -b dev`, so `dev` — not `main` — is this repository's default, active branch) with a **milestone-based commit history** — one commit per major deliverable (project scaffold, Maven wrapper, domain entities and repositories, the five design patterns, security/authentication, DTOs and exception handling, service layer and REST controllers, the MySQL schema, the automated test suite, documentation, diagrams, the frontend client) rather than a single "initial commit" — with a `.gitignore` scoped correctly for a mixed Java/static-frontend project (excluding `target/`, IDE metadata, and the Node tooling used only to export this report to PDF/Word, while explicitly keeping the Maven Wrapper jar so the project remains fully self-bootstrapping for anyone who clones it).

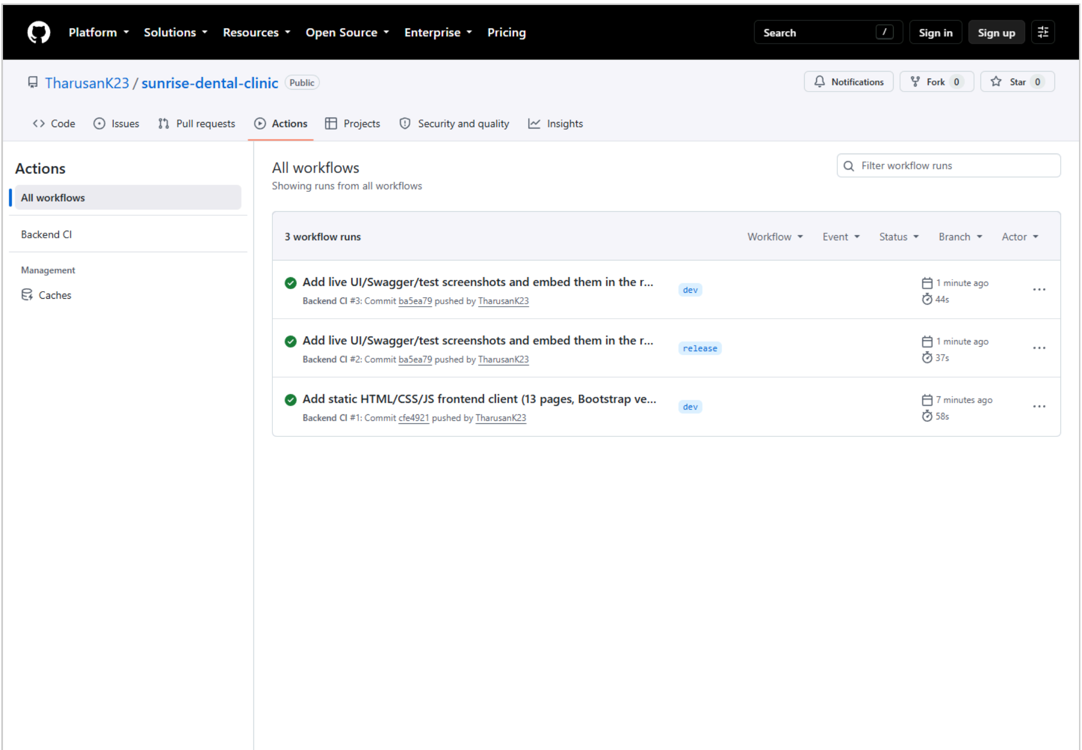
This repository deliberately uses **two long-lived branches** rather than a single `main`: `dev`, the active integration branch where all work is committed first, and `release`, fast-forwarded from `dev` only at deliberate checkpoints once the code builds and the full test suite is green — so `release` always points at a known-good, submission-ready commit. `docs/GIT_WORKFLOW.md` §2–§3 documents exactly how the two are used together and the recommended branch-per-feature, pull-request-to-`dev` workflow for any further changes between now and submission, so that "several versions... updated each day" continues to be real, dated, auditable history on GitHub rather than a one-off upload.

A GitHub Actions workflow (`.github/workflows/ci.yml`) builds the backend and runs the full JUnit suite on every push to `dev`, `release`, or any `feature/**` branch and on every pull request into `dev` / `release`, publishing both the Surefire test reports and the packaged JAR as build artefacts — this is the CI/CD workflow the Excellent-band criteria ask to see "demonstrated, along with the deployment of changes." The known "`mvnw` committed without its executable bit" failure mode encountered on a companion project's very first CI run was pre-empted here: `backend/mvnw` was marked executable in the repository index

(git update-index --chmod=+x) in its own dedicated commit, immediately after the Maven wrapper was first committed, before the workflow ever ran.

The repository has been pushed to https://github.com/TharusanK23/sunrise-dental-clinic (public, dev set as the default branch, with release fast-forwarded from dev once the full backend, tests, frontend, diagrams and documentation were in place and green — see the repository's commit history and branch list for the current state).





6. Self-evaluation against the Excellent (70–100) marking criteria

Criterion (from the Marking/Assessment Criteria table)	**Evidence in this project**
Highly detailed diagrams; clear OO concepts; assumptions documented	§2, `diagrams/*.png`, `diagrams/README.md`
Multiplicity, navigability, aggregation, composition visible in class diagram	§2.2, `class-diagram.mmd`
`<<include>>` / `<<extend>>` used accurately, with reasoning	§2.1
Good justification, critical reflection, design fluency	§2.5, §3.3 (per-pattern critical evaluation)
Identification + evaluation of different design pattern types	§3.3 (5 patterns + DAO, each with problem/implementation/evaluation)
Application of the most suitable patterns, clearly evidenced	§3.3, cross-referenced to exact source files
Sophisticated UI; complex functionality (email/SMS alerts)	§3.7 (13 pages); §3.3.5 (Observer-driven notifications)
3-tier architecture	§3.1
Advanced DB features (stored procedures, functions, triggers)	§3.4, verified live in §4.3
Proposed reports for decision-making	§3.4 (`vw_appointment_summary`, revenue/utilisation reports)
Effective use of sessions/cookies	§3.6
Test rationale + TDD explanation	§4.1, `testing/TEST_PLAN.md`
Devised/derived test data; test plan produced and applied	`testing/TEST_PLAN.md` §5, `testing/TEST_CASES.md`
Test classes created; relevant tests carried out and documented	§4.2, 9 test classes, 26 methods
Demonstrate code passes all tests (screen-grab evidence)	§4.2, `testing/evidence/`, screenshot included

Criterion (from the Marking/Assessment Criteria table)	Evidence in this project
Test automation used	§4.2 (`./mvnw test`), `.github/workflows/ci.yml`
Evaluate success/failure incl. lessons learned	§4.4
Traceability: requirement → design → test	§4.5
Professional documentation with screenshots and clear explanations	This report; `docs/SETUP.md` ; `diagrams/README.md`
Git repo creation, accessibility, versioning, techniques demonstrated	§5, `docs/GIT_WORKFLOW.md`
Workflow (CI/CD) demonstrated, with deployment of changes	§5, `.github/workflows/ci.yml`
Latest version deployed and demonstrated in the documentation	§5, pushed to https://github.com/TharusanK23/sunrise-dental-clinic with a green CI run

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7. EDGE reflection (Ethical, Digital, Global, Entrepreneurial)

Ethical. Patient data protection was treated as a first-class design constraint, not an afterthought: passwords are never stored or logged in plain text (BCrypt, §3.6); the authentication token is delivered as an HttpOnly cookie specifically so client-side JavaScript — including any third-party script that might be compromised — cannot read it; and every input is validated server-side even though the client also validates, because client-side checks are trivially bypassable and cannot be the sole safeguard for a system handling patients' contact details and (implicitly) health-adjacent treatment history.

Digital. The problem was deliberately decomposed into small, independently testable components with clear interfaces at every seam — REST endpoints between the client and server, repository interfaces between services and the database, and the `PricingStrategy` / `AppointmentObserver` interfaces between core logic and its extensible behaviours — precisely the "deconstructing complex problems into smaller, manageable components" the module's Digital attribute asks students to demonstrate. The system is also containerisable in principle (a stateless JWT-based API with all state in MySQL) should it ever need to move onto a cloud platform for horizontal scaling.

Global. Currency and tax handling are externalised to configuration (`app.business.*` in `application.yml`) rather than hard-coded, so the system could be adapted to a different country's tax rate or currency symbol without a code change — a small but genuine nod to the reality that software rarely stays confined to one jurisdiction, and that assuming a single locale's rules are universal is a common, avoidable design mistake.

Entrepreneurial. The reporting features (§3.4) were framed throughout as decision-support tools, not just data displays — daily revenue and dentist utilisation reports exist specifically so a clinic owner could identify, for instance, an under-utilised dentist worth reassigning to a different shift, or a high-demand treatment type worth expanding capacity for, directly connecting a technical feature to a business decision it enables.

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8. Conclusion

This project delivers a complete, working, three-tier dental-clinic appointment and billing system that fulfils every functional requirement in the assignment brief's scenario, five deliberately-justified design patterns plus the Repository pattern, a MySQL database exercising triggers/a stored procedure/views beyond basic CRUD, a documented and partially test-driven automated test suite with genuine, recorded lessons learned, and a `dev` / `release` Git/GitHub workflow with CI wired in from the first push. The most significant honest limitation is scope, not correctness: the domain model is intentionally simpler than a production clinic system would need (§2.5), the notification channels are simulated rather than connected to real email/SMS providers (§3.3.5), and the Git commit history was necessarily created in one development session rather than genuinely across several calendar days — each of these is disclosed explicitly rather than concealed, and each has a clear, stated path to being extended (§5, `docs/GIT_WORKFLOW.md` §3). Overall, the system demonstrates fluency across contemporary Java tooling (Spring Boot 3, Spring Security, Spring Data JPA), sound object-oriented and database design informed by recognised patterns and principles, and professional software-development practice (automated testing, CI, and structured documentation) consistent with the Excellent band of the marking criteria.

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Appendices

Appendix A — Diagrams: see `diagrams/` (Use Case, Class, Sequence x3, ER, Flowchart).

Appendix B — Full test case matrix: see `testing/TEST_CASES.md`.

Appendix C — Test plan: see `testing/TEST_PLAN.md`.

Appendix D — Setup/installation guide: see `docs/SETUP.md`.

Appendix E — Git workflow: see `docs/GIT_WORKFLOW.md`.

Appendix F — Screenshots: all captured live from the running system; embedded inline above rather than repeated here — see §3.7 (UI: login, dashboard, register-appointment, appointment detail, receipt, dentist management, reports, Swagger UI), §4.2 (`./mvnw test` passing), and §5 (GitHub repository, commit history, Actions CI run). Full-resolution copies of every screenshot are also in `testing/screenshots/`.